Genuine Multipartite Entanglement Analysis

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Model                              Ground-state properties           GME summary
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Hubbard ring (1-D, PBC)            E0                : -4.60352630  Extracted n-spin RDM : n = 8
                                   E1                : -4.29999276
System parameters                                                   Purity:
-----------------                  <S2>              : -0.000000      gamma = Tr(rho^2)
Sites (L)          : 8             S                 : -0.0000
Electrons (Ne)     : 8             <Sz>              : 0.000000     Separable cut criterion:
Hopping (t)        : 1.0           Multiplicity 2S+1 : 1.0            gamma_A = 1  <->  rho = rho_A x rho_B
Interaction (U)    : 4.0           Spin state        : Singlet (S=0)
Mode               : uniform                                       C_GME (Troiani et al.):
n-body order       : 8             Double occupancy                   C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))
                                   ----------------
                                   <D> total         : 0.759406     Convex roof (mixed):
                                   <D> / site        : 0.094926       C_GME(ρ) = min Σ pᵢ C_GME(|ψᵢ⟩)

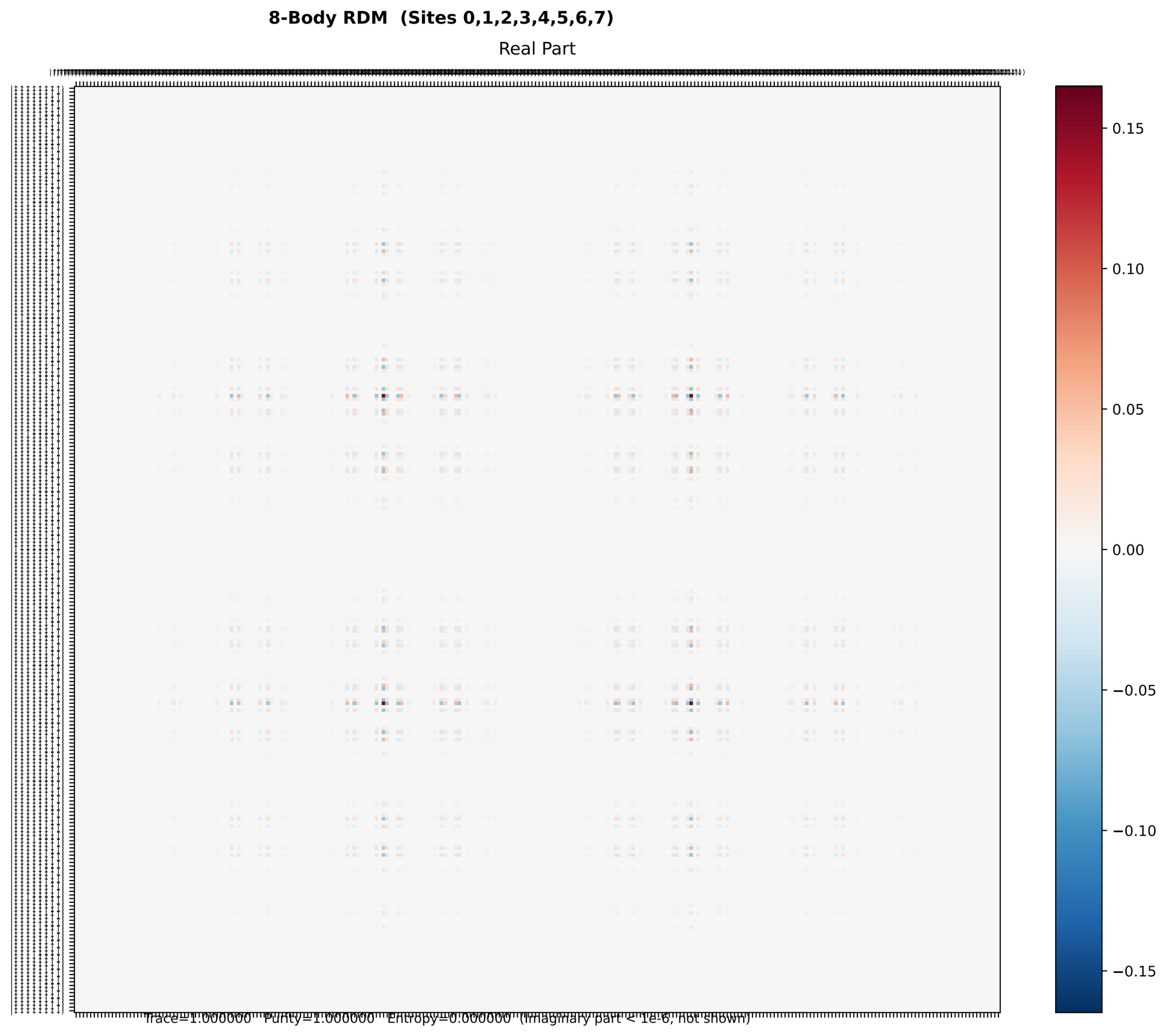

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Summary: All 8-Body RDM Clusters

System: L=8, Ne=8, U=4.0, t=1.0 | E_0=-4.603526

Sites	Purity	C_GME	C_LB	C_cayl	C_expo	C_UB	Status
(0, 1, 2, 3, 4, 5,	1.000000	0.971943	0.313942	0.971943	0.971943	0.971943	GME
(1, 2, 3, 4, 5, 6,	1.000000	0.971943	0.313942	0.971943	0.971943	0.971943	GME
(2, 3, 4, 5, 6, 7,	1.000000	0.971943	0.313942	0.971943	0.971943	0.971943	GME
(3, 4, 5, 6, 7, 0,	1.000000	0.971943	0.313942	0.971943	0.971943	0.971943	GME
(4, 5, 6, 7, 0, 1,	1.000000	0.971943	0.313942	0.971943	0.971943	0.971943	GME
(5, 6, 7, 0, 1, 2,	1.000000	0.971943	0.313942	0.971943	0.971943	0.971943	GME
(6, 7, 0, 1, 2, 3,	1.000000	0.971943	0.313942	0.971943	0.971943	0.971943	GME
(7, 0, 1, 2, 3, 4,	1.000000	0.971943	0.313942	0.971943	0.971943	0.971943	GME

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state


8-Body RDM (Sites 0,1,2,3,4,5,6,7)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 8-Body RDM (Sites 0,1,2,3,4,5,6,7)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000001
Eigenvalue min	-3.324230e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

Convex Roof Analysis
==========

Lower Bounds:
- Paper (witness) = 0.00000000
- Notes (6-9) = 0.31394240
- Best C_LB = 0.31394240

C_GME (unitary optimization - Cayley Transform):
- C_num (unitary opt) = 0.97194283 BEST
- (20 restarts)

C_GME (unitary optimization - Expoenetial Map):
- C_num = 0.97194283 BEST
- (L-BFGS-B, 20 restarts)

Upper Bound:
- C_UB (eigendecomp) = 0.97194283

Status: GME
Bounds consistent: YES
Convergence: TIGHT

GME Analysis
==========

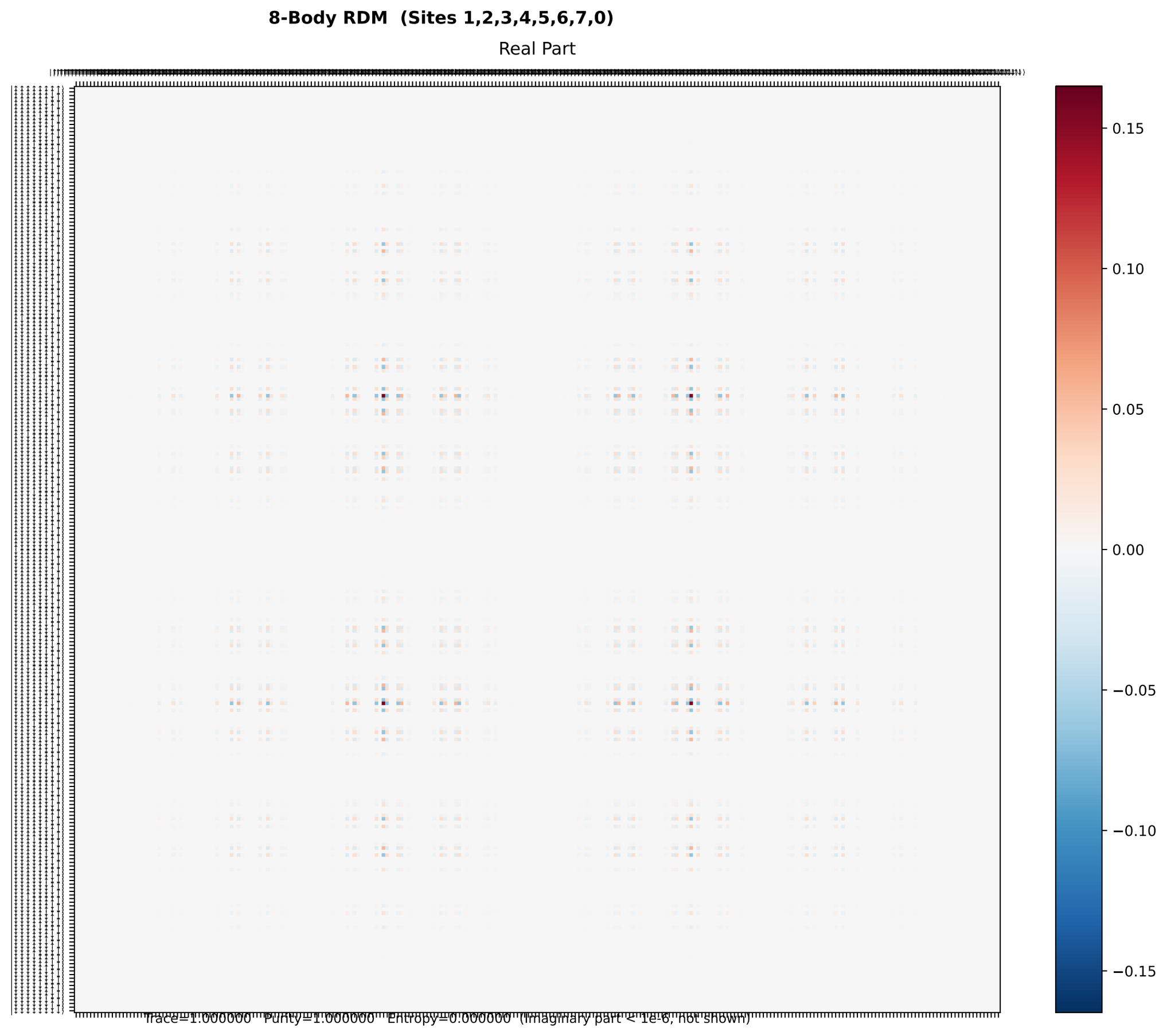
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(0,)	(1, 2, 3, 4, 5, 6, 7)	0.50000000
(0, 1)	(2, 3, 4, 5, 6, 7)	0.52766357
(0, 2)	(1, 3, 4, 5, 6, 7)	0.29980473
(0, 3)	(1, 2, 4, 5, 6, 7)	0.29544345
(0, 4)	(1, 2, 3, 5, 6, 7)	0.27817710
(0, 5)	(1, 2, 3, 4, 6, 7)	0.29544345
(0, 6)	(1, 2, 3, 4, 5, 7)	0.29980473
(0, 7)	(1, 2, 3, 4, 5, 6)	0.52766357
(0, 1, 2)	(3, 4, 5, 6, 7)	0.42756594
(0, 1, 3)	(2, 4, 5, 6, 7)	0.31145588
(0, 1, 4)	(2, 3, 5, 6, 7)	0.30064206
(0, 1, 5)	(2, 3, 4, 6, 7)	0.30064206
(0, 1, 6)	(2, 3, 4, 5, 7)	0.31145588
(0, 1, 7)	(2, 3, 4, 5, 6)	0.42756594
(0, 2, 3)	(1, 4, 5, 6, 7)	0.31145588
(0, 2, 4)	(1, 3, 5, 6, 7)	0.18889328
(0, 2, 5)	(1, 3, 4, 6, 7)	0.19534582
(0, 2, 6)	(1, 3, 4, 5, 7)	0.18889328
(0, 2, 7)	(1, 3, 4, 5, 6)	0.31145588
(0, 3, 4)	(1, 2, 5, 6, 7)	0.30064206
(0, 3, 5)	(1, 2, 4, 6, 7)	0.19534582
(0, 3, 6)	(1, 2, 4, 5, 7)	0.19534582
(0, 3, 7)	(1, 2, 4, 5, 6)	0.30064206
(0, 4, 5)	(1, 2, 3, 6, 7)	0.30064206
(0, 4, 6)	(1, 2, 3, 5, 7)	0.18889328
(0, 4, 7)	(1, 2, 3, 5, 6)	0.30064206
(0, 5, 6)	(1, 2, 3, 4, 7)	0.31145588
(0, 5, 7)	(1, 2, 3, 4, 6)	0.31145588
(0, 6, 7)	(1, 2, 3, 4, 5)	0.42756594
(0, 1, 2, 3)	(4, 5, 6, 7)	0.47865643
(0, 1, 2, 4)	(3, 5, 6, 7)	0.27204245
(0, 1, 2, 5)	(3, 4, 6, 7)	0.29330035
(0, 1, 2, 6)	(3, 4, 5, 7)	0.27204245
(0, 1, 2, 7)	(3, 4, 5, 6)	0.47865643
(0, 1, 3, 4)	(2, 5, 6, 7)	0.29330035
(0, 1, 3, 5)	(2, 4, 6, 7)	0.20203136
(0, 1, 3, 6)	(2, 4, 5, 7)	0.20033735
(0, 1, 3, 7)	(2, 4, 5, 6)	0.27204245
(0, 1, 4, 5)	(2, 3, 6, 7)	0.35325200
(0, 1, 4, 6)	(2, 3, 5, 7)	0.20203136
(0, 1, 4, 7)	(2, 3, 5, 6)	0.29330035
(0, 1, 5, 6)	(2, 3, 4, 7)	0.29330035
(0, 1, 5, 7)	(2, 3, 4, 6)	0.27204245
(0, 1, 6, 7)	(2, 3, 4, 5)	0.47865643
(0, 2, 3, 4)	(1, 5, 6, 7)	0.27204245
(0, 2, 3, 5)	(1, 4, 6, 7)	0.20033735
(0, 2, 3, 6)	(1, 4, 5, 7)	0.20203136
(0, 2, 3, 7)	(1, 4, 5, 6)	0.29330035
(0, 2, 4, 5)	(1, 3, 6, 7)	0.20203136
(0, 2, 4, 6)	(1, 3, 5, 7)	0.13053255
(0, 2, 4, 7)	(1, 3, 5, 6)	0.20203136
(0, 2, 5, 6)	(1, 3, 4, 7)	0.20203136
(0, 2, 5, 7)	(1, 3, 4, 6)	0.20033735
(0, 2, 6, 7)	(1, 3, 4, 5)	0.27204245
(0, 3, 4, 5)	(1, 2, 6, 7)	0.29330035
(0, 3, 4, 6)	(1, 2, 5, 7)	0.20203136
(0, 3, 4, 7)	(1, 2, 5, 6)	0.35325200
(0, 3, 5, 6)	(1, 2, 4, 7)	0.20033735
(0, 3, 5, 7)	(1, 2, 4, 6)	0.20203136
(0, 3, 6, 7)	(1, 2, 4, 5)	0.29330035
(0, 4, 5, 6)	(1, 2, 3, 7)	0.27204245
(0, 4, 5, 7)	(1, 2, 3, 6)	0.29330035
(0, 4, 6, 7)	(1, 2, 3, 5)	0.27204245
(0, 5, 6, 7)	(1, 2, 3, 4)	0.47865643
(0, 1, 2, 3, 4)	(5, 6, 7)	0.42756594
(0, 1, 2, 3, 5)	(4, 6, 7)	0.31145588
(0, 1, 2, 3, 6)	(4, 5, 7)	0.31145588
(0, 1, 2, 3, 7)	(4, 5, 6)	0.42756594
(0, 1, 2, 4, 5)	(3, 6, 7)	0.30064206
(0, 1, 2, 4, 6)	(3, 5, 7)	0.18889328
(0, 1, 2, 4, 7)	(3, 5, 6)	0.31145588
(0, 1, 2, 5, 6)	(3, 4, 7)	0.30064206
(0, 1, 2, 5, 7)	(3, 4, 6)	0.31145588
(0, 1, 2, 6, 7)	(3, 4, 5)	0.42756594
(0, 1, 3, 4, 5)	(2, 6, 7)	0.30064206
(0, 1, 3, 4, 6)	(2, 5, 7)	0.19534582
(0, 1, 3, 4, 7)	(2, 5, 6)	0.30064206
(0, 1, 3, 5, 6)	(2, 4, 7)	0.19534582
(0, 1, 3, 5, 7)	(2, 4, 6)	0.18889328
(0, 1, 3, 6, 7)	(2, 4, 5)	0.31145588
(0, 1, 4, 5, 6)	(2, 3, 7)	0.30064206
(0, 1, 4, 5, 7)	(2, 3, 6)	0.30064206
(0, 1, 4, 6, 7)	(2, 3, 5)	0.31145588
(0, 1, 5, 6, 7)	(2, 3, 4)	0.42756594
(0, 2, 3, 4, 5)	(1, 6, 7)	0.31145588
(0, 2, 3, 4, 6)	(1, 5, 7)	0.18889328
(0, 2, 3, 4, 7)	(1, 5, 6)	0.30064206
(0, 2, 3, 5, 6)	(1, 4, 7)	0.19534582
(0, 2, 3, 5, 7)	(1, 4, 6)	0.19534582
(0, 2, 3, 6, 7)	(1, 4, 5)	0.30064206
(0, 2, 4, 5, 6)	(1, 3, 7)	0.18889328
(0, 2, 4, 5, 7)	(1, 3, 6)	0.19534582
(0, 2, 4, 6, 7)	(1, 3, 5)	0.18889328
(0, 2, 5, 6, 7)	(1, 3, 4)	0.31145588
(0, 3, 4, 5, 6)	(1, 2, 7)	0.31145588
(0, 3, 4, 5, 7)	(1, 2, 6)	0.30064206
(0, 3, 4, 6, 7)	(1, 2, 5)	0.30064206
(0, 3, 5, 6, 7)	(1, 2, 4)	0.31145588
(0, 4, 5, 6, 7)	(1, 2, 3)	0.42756594
(0, 1, 2, 3, 4, 5)	(6, 7)	0.52766357
(0, 1, 2, 3, 4, 6)	(5, 7)	0.29980473
(0, 1, 2, 3, 4, 7)	(5, 6)	0.52766357
(0, 1, 2, 3, 5, 6)	(4, 7)	0.29544345
(0, 1, 2, 3, 5, 7)	(4, 6)	0.29980473
(0, 1, 2, 3, 6, 7)	(4, 5)	0.52766357
(0, 1, 2, 4, 5, 6)	(3, 7)	0.27817710
(0, 1, 2, 4, 5, 7)	(3, 6)	0.29544345
(0, 1, 2, 4, 6, 7)	(3, 5)	0.29980473
(0, 1, 2, 5, 6, 7)	(3, 4)	0.52766357
(0, 1, 3, 4, 5, 6)	(2, 7)	0.29544345
(0, 1, 3, 4, 5, 7)	(2, 6)	0.27817710
(0, 1, 3, 4, 6, 7)	(2, 5)	0.29544345
(0, 1, 3, 5, 6, 7)	(2, 4)	0.29980473
(0, 1, 4, 5, 6, 7)	(2, 3)	0.52766357
(0, 2, 3, 4, 5, 6)	(1, 7)	0.29980473
(0, 2, 3, 4, 5, 7)	(1, 6)	0.29544345
(0, 2, 3, 4, 6, 7)	(1, 5)	0.27817710
(0, 2, 3, 5, 6, 7)	(1, 4)	0.29544345
(0, 2, 4, 5, 6, 7)	(1, 3)	0.29980473
(0, 3, 4, 5, 6, 7)	(1, 2)	0.52766357
(0, 1, 2, 3, 4, 5, 6)	(7,)	0.50000000
(0, 1, 2, 3, 4, 5, 7)	(6,)	0.50000000
(0, 1, 2, 3, 4, 6, 7)	(5,)	0.50000000
(0, 1, 2, 3, 5, 6, 7)	(4,)	0.50000000
(0, 1, 2, 4, 5, 6, 7)	(3,)	0.50000000
(0, 1, 3, 4, 5, 6, 7)	(2,)	0.50000000
(0, 2, 3, 4, 5, 6, 7)	(1,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.97194283
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

C_exact = 0.97194283
C_LB = 0.31394240
Gap = 0.65800042


8-Body RDM (Sites 1,2,3,4,5,6,7,0)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 8-Body RDM (Sites 1,2,3,4,5,6,7,0)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000001
Eigenvalue min	-3.355016e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

Convex Roof Analysis
==========

Lower Bounds:
- Paper (witness) = 0.00000000
- Notes (6-9) = 0.31394240
- Best C_LB = 0.31394240

C_GME (unitary optimization - Cayley Transform):
- C_num (unitary opt) = 0.97194283 BEST
- (20 restarts)

C_GME (unitary optimization - Expoenetial Map):
- C_num = 0.97194283 BEST
- (L-BFGS-B, 20 restarts)

Upper Bound:
- C_UB (eigendecomp) = 0.97194283

Status: GME
Bounds consistent: YES
Convergence: TIGHT

GME Analysis
==========

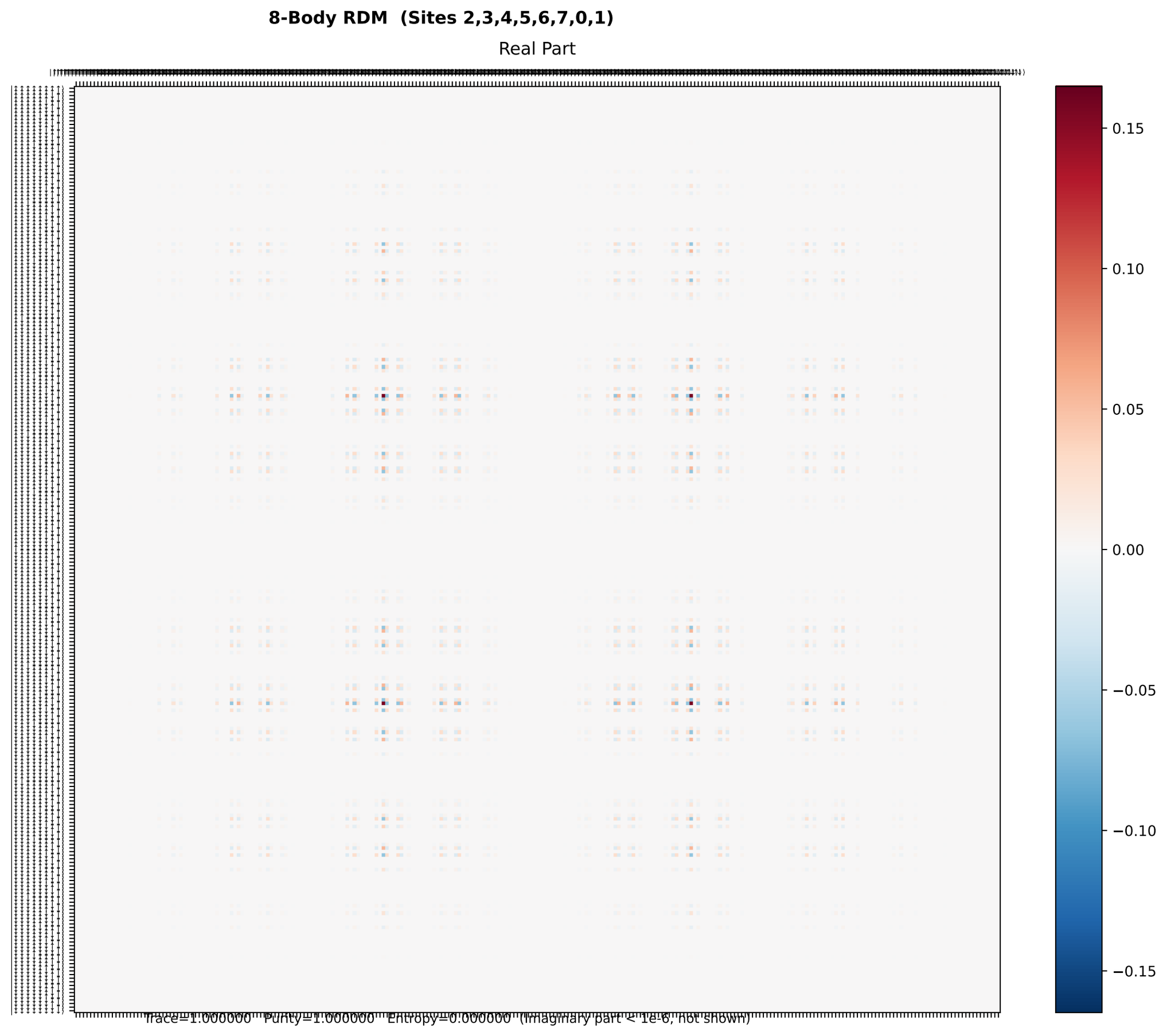
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(1,)	(2, 3, 4, 5, 6, 7, 0)	0.50000000
(1, 2)	(3, 4, 5, 6, 7, 0)	0.52766357
(1, 3)	(2, 4, 5, 6, 7, 0)	0.29980473
(1, 4)	(2, 3, 5, 6, 7, 0)	0.29544345
(1, 5)	(2, 3, 4, 6, 7, 0)	0.27817710
(1, 6)	(2, 3, 4, 5, 7, 0)	0.29544345
(1, 7)	(2, 3, 4, 5, 6, 0)	0.29980473
(1, 0)	(2, 3, 4, 5, 6, 7)	0.52766357
(1, 2, 3)	(4, 5, 6, 7, 0)	0.42756594
(1, 2, 4)	(3, 5, 6, 7, 0)	0.31145588
(1, 2, 5)	(3, 4, 6, 7, 0)	0.30064206
(1, 2, 6)	(3, 4, 5, 7, 0)	0.30064206
(1, 2, 7)	(3, 4, 5, 6, 0)	0.31145588
(1, 2, 0)	(3, 4, 5, 6, 7)	0.42756594
(1, 3, 4)	(2, 5, 6, 7, 0)	0.31145588
(1, 3, 5)	(2, 4, 6, 7, 0)	0.18889328
(1, 3, 6)	(2, 4, 5, 7, 0)	0.19534582
(1, 3, 7)	(2, 4, 5, 6, 0)	0.18889328
(1, 3, 0)	(2, 4, 5, 6, 7)	0.31145588
(1, 4, 5)	(2, 3, 6, 7, 0)	0.30064206
(1, 4, 6)	(2, 3, 5, 7, 0)	0.19534582
(1, 4, 7)	(2, 3, 5, 6, 0)	0.19534582
(1, 4, 0)	(2, 3, 5, 6, 7)	0.30064206
(1, 5, 6)	(2, 3, 4, 7, 0)	0.30064206
(1, 5, 7)	(2, 3, 4, 6, 0)	0.18889328
(1, 5, 0)	(2, 3, 4, 6, 7)	0.30064206
(1, 6, 7)	(2, 3, 4, 5, 0)	0.31145588
(1, 6, 0)	(2, 3, 4, 5, 7)	0.31145588
(1, 7, 0)	(2, 3, 4, 5, 6)	0.42756594
(1, 2, 3, 4)	(5, 6, 7, 0)	0.47865643
(1, 2, 3, 5)	(4, 6, 7, 0)	0.27204245
(1, 2, 3, 6)	(4, 5, 7, 0)	0.29330035
(1, 2, 3, 7)	(4, 5, 6, 0)	0.27204245
(1, 2, 3, 0)	(4, 5, 6, 7)	0.47865643
(1, 2, 4, 5)	(3, 6, 7, 0)	0.29330035
(1, 2, 4, 6)	(3, 5, 7, 0)	0.20203136
(1, 2, 4, 7)	(3, 5, 6, 0)	0.20033735
(1, 2, 4, 0)	(3, 5, 6, 7)	0.27204245
(1, 2, 5, 6)	(3, 4, 7, 0)	0.35325200
(1, 2, 5, 7)	(3, 4, 6, 0)	0.20203136
(1, 2, 5, 0)	(3, 4, 6, 7)	0.29330035
(1, 2, 6, 7)	(3, 4, 5, 0)	0.29330035
(1, 2, 6, 0)	(3, 4, 5, 7)	0.27204245
(1, 2, 7, 0)	(3, 4, 5, 6)	0.47865643
(1, 3, 4, 5)	(2, 6, 7, 0)	0.27204245
(1, 3, 4, 6)	(2, 5, 7, 0)	0.20033735
(1, 3, 4, 7)	(2, 5, 6, 0)	0.20203136
(1, 3, 4, 0)	(2, 5, 6, 7)	0.29330035
(1, 3, 5, 6)	(2, 4, 7, 0)	0.20203136
(1, 3, 5, 7)	(2, 4, 6, 0)	0.13053255
(1, 3, 5, 0)	(2, 4, 6, 7)	0.20203136
(1, 3, 6, 7)	(2, 4, 5, 0)	0.20203136
(1, 3, 6, 0)	(2, 4, 5, 7)	0.20033735
(1, 3, 7, 0)	(2, 4, 5, 6)	0.27204245
(1, 4, 5, 6)	(2, 3, 7, 0)	0.29330035
(1, 4, 5, 7)	(2, 3, 6, 0)	0.20203136
(1, 4, 5, 0)	(2, 3, 6, 7)	0.35325200
(1, 4, 6, 7)	(2, 3, 5, 0)	0.20033735
(1, 4, 6, 0)	(2, 3, 5, 7)	0.20203136
(1, 4, 7, 0)	(2, 3, 5, 6)	0.29330035
(1, 5, 6, 7)	(2, 3, 4, 0)	0.27204245
(1, 5, 6, 0)	(2, 3, 4, 7)	0.29330035
(1, 5, 7, 0)	(2, 3, 4, 6)	0.27204245
(1, 6, 7, 0)	(2, 3, 4, 5)	0.47865643
(1, 2, 3, 4, 5)	(6, 7, 0)	0.42756594
(1, 2, 3, 4, 6)	(5, 7, 0)	0.31145588
(1, 2, 3, 4, 7)	(5, 6, 0)	0.31145588
(1, 2, 3, 4, 0)	(5, 6, 7)	0.42756594
(1, 2, 3, 5, 6)	(4, 7, 0)	0.30064206
(1, 2, 3, 5, 7)	(4, 6, 0)	0.18889328
(1, 2, 3, 5, 0)	(4, 6, 7)	0.31145588
(1, 2, 3, 6, 7)	(4, 5, 0)	0.30064206
(1, 2, 3, 6, 0)	(4, 5, 7)	0.31145588
(1, 2, 3, 7, 0)	(4, 5, 6)	0.42756594
(1, 2, 4, 5, 6)	(3, 7, 0)	0.30064206
(1, 2, 4, 5, 7)	(3, 6, 0)	0.19534582
(1, 2, 4, 5, 0)	(3, 6, 7)	0.30064206
(1, 2, 4, 6, 7)	(3, 5, 0)	0.19534582
(1, 2, 4, 6, 0)	(3, 5, 7)	0.18889328
(1, 2, 4, 7, 0)	(3, 5, 6)	0.31145588
(1, 2, 5, 6, 7)	(3, 4, 0)	0.30064206
(1, 2, 5, 6, 0)	(3, 4, 7)	0.30064206
(1, 2, 5, 7, 0)	(3, 4, 6)	0.31145588
(1, 2, 6, 7, 0)	(3, 4, 5)	0.42756594
(1, 3, 4, 5, 6)	(2, 7, 0)	0.31145588
(1, 3, 4, 5, 7)	(2, 6, 0)	0.18889328
(1, 3, 4, 5, 0)	(2, 6, 7)	0.30064206
(1, 3, 4, 6, 7)	(2, 5, 0)	0.19534582
(1, 3, 4, 6, 0)	(2, 5, 7)	0.19534582
(1, 3, 4, 7, 0)	(2, 5, 6)	0.30064206
(1, 3, 5, 6, 7)	(2, 4, 0)	0.18889328
(1, 3, 5, 6, 0)	(2, 4, 7)	0.19534582
(1, 3, 5, 7, 0)	(2, 4, 6)	0.18889328
(1, 3, 6, 7, 0)	(2, 4, 5)	0.31145588
(1, 4, 5, 6, 7)	(2, 3, 0)	0.31145588
(1, 4, 5, 6, 0)	(2, 3, 7)	0.30064206
(1, 4, 5, 7, 0)	(2, 3, 6)	0.30064206
(1, 4, 6, 7, 0)	(2, 3, 5)	0.31145588
(1, 5, 6, 7, 0)	(2, 3, 4)	0.42756594
(1, 2, 3, 4, 5, 6)	(7, 0)	0.52766357
(1, 2, 3, 4, 5, 7)	(6, 0)	0.29980473
(1, 2, 3, 4, 5, 0)	(6, 7)	0.52766357
(1, 2, 3, 4, 6, 7)	(5, 0)	0.29544345
(1, 2, 3, 4, 6, 0)	(5, 7)	0.29980473
(1, 2, 3, 4, 7, 0)	(5, 6)	0.52766357
(1, 2, 3, 5, 6, 7)	(4, 0)	0.27817710
(1, 2, 3, 5, 6, 0)	(4, 7)	0.29544345
(1, 2, 3, 5, 7, 0)	(4, 6)	0.29980473
(1, 2, 3, 6, 7, 0)	(4, 5)	0.52766357
(1, 2, 4, 5, 6, 7)	(3, 0)	0.29544345
(1, 2, 4, 5, 6, 0)	(3, 7)	0.27817710
(1, 2, 4, 5, 7, 0)	(3, 6)	0.29544345
(1, 2, 4, 6, 7, 0)	(3, 5)	0.29980473
(1, 2, 5, 6, 7, 0)	(3, 4)	0.52766357
(1, 3, 4, 5, 6, 7)	(2, 0)	0.29980473
(1, 3, 4, 5, 6, 0)	(2, 7)	0.29544345
(1, 3, 4, 5, 7, 0)	(2, 6)	0.27817710
(1, 3, 4, 6, 7, 0)	(2, 5)	0.29544345
(1, 3, 5, 6, 7, 0)	(2, 4)	0.29980473
(1, 4, 5, 6, 7, 0)	(2, 3)	0.52766357
(1, 2, 3, 4, 5, 6, 7)	(0,)	0.50000000
(1, 2, 3, 4, 5, 6, 0)	(7,)	0.50000000
(1, 2, 3, 4, 5, 7, 0)	(6,)	0.50000000
(1, 2, 3, 4, 6, 7, 0)	(5,)	0.50000000
(1, 2, 3, 5, 6, 7, 0)	(4,)	0.50000000
(1, 2, 4, 5, 6, 7, 0)	(3,)	0.50000000
(1, 3, 4, 5, 6, 7, 0)	(2,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.97194283
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

- C_exact = 0.97194283
- C_LB = 0.31394240
- Gap = 0.65800042


8-Body RDM (Sites 2,3,4,5,6,7,0,1)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 8-Body RDM (Sites 2,3,4,5,6,7,0,1)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000001
Eigenvalue min	-3.613130e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

Convex Roof Analysis
==========

Lower Bounds:
- Paper (witness) = 0.00000000
- Notes (6-9) = 0.31394240
- Best C_LB = 0.31394240

C_GME (unitary optimization - Cayley Transform):
- C_num (unitary opt) = 0.97194283 BEST
- (20 restarts)

C_GME (unitary optimization - Expoenetial Map):
- C_num = 0.97194283 BEST
- (L-BFGS-B, 20 restarts)

Upper Bound:
- C_UB (eigendecomp) = 0.97194283

Status: GME
Bounds consistent: YES
Convergence: TIGHT

GME Analysis
==========

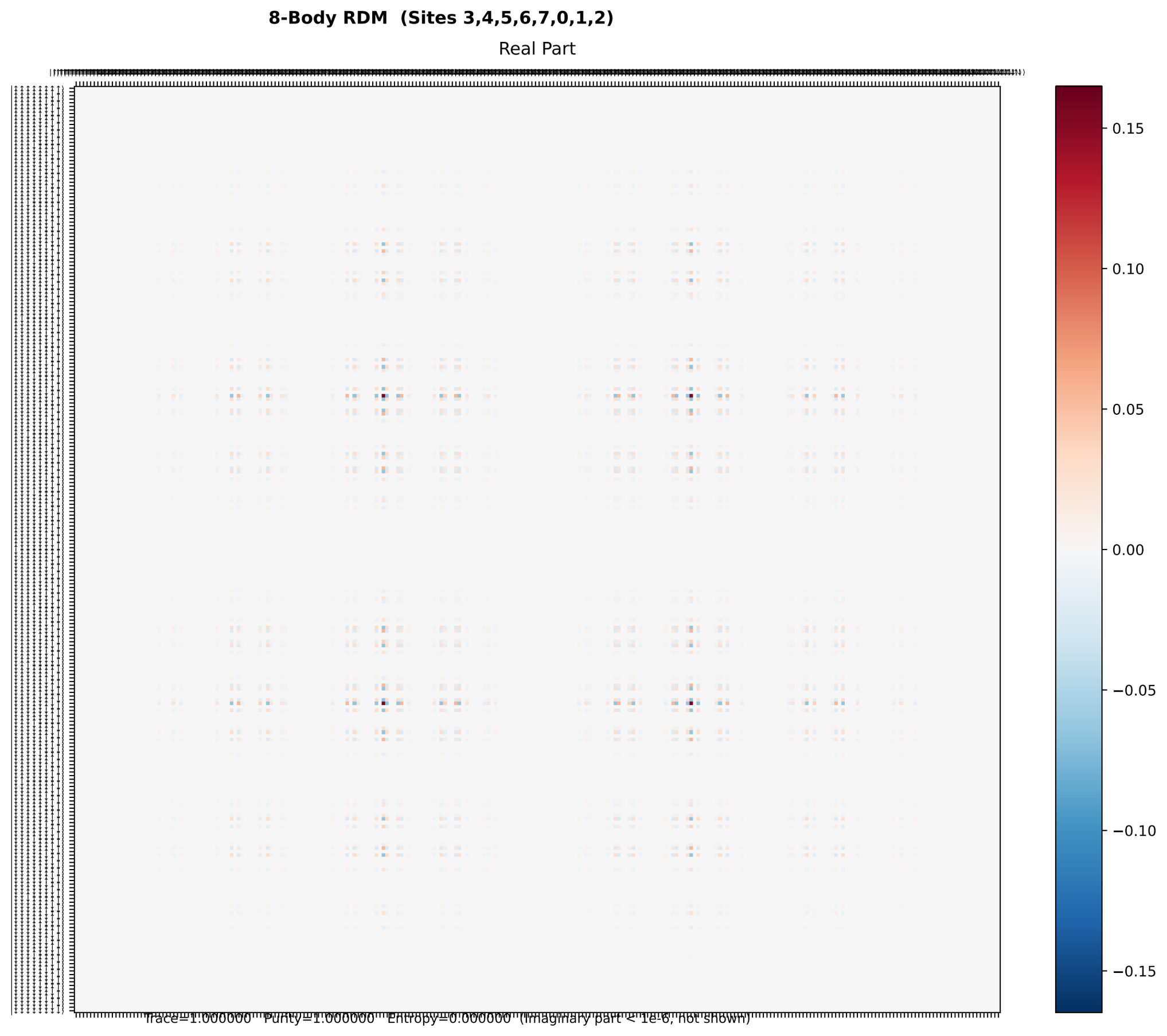
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(2,)	(3, 4, 5, 6, 7, 0, 1)	0.50000000
(2, 3)	(4, 5, 6, 7, 0, 1)	0.52766357
(2, 4)	(3, 5, 6, 7, 0, 1)	0.29980473
(2, 5)	(3, 4, 6, 7, 0, 1)	0.29544345
(2, 6)	(3, 4, 5, 7, 0, 1)	0.27817710
(2, 7)	(3, 4, 5, 6, 0, 1)	0.29544345
(2, 0)	(3, 4, 5, 6, 7, 1)	0.29980473
(2, 1)	(3, 4, 5, 6, 7, 0)	0.52766357
(2, 3, 4)	(5, 6, 7, 0, 1)	0.42756594
(2, 3, 5)	(4, 6, 7, 0, 1)	0.31145588
(2, 3, 6)	(4, 5, 7, 0, 1)	0.30064206
(2, 3, 7)	(4, 5, 6, 0, 1)	0.30064206
(2, 3, 0)	(4, 5, 6, 7, 1)	0.31145588
(2, 3, 1)	(4, 5, 6, 7, 0)	0.42756594
(2, 4, 5)	(3, 6, 7, 0, 1)	0.31145588
(2, 4, 6)	(3, 5, 7, 0, 1)	0.18889328
(2, 4, 7)	(3, 5, 6, 0, 1)	0.19534582
(2, 4, 0)	(3, 5, 6, 7, 1)	0.18889328
(2, 4, 1)	(3, 5, 6, 7, 0)	0.31145588
(2, 5, 6)	(3, 4, 7, 0, 1)	0.30064206
(2, 5, 7)	(3, 4, 6, 0, 1)	0.19534582
(2, 5, 0)	(3, 4, 6, 7, 1)	0.19534582
(2, 5, 1)	(3, 4, 6, 7, 0)	0.30064206
(2, 6, 7)	(3, 4, 5, 0, 1)	0.30064206
(2, 6, 0)	(3, 4, 5, 7, 1)	0.18889328
(2, 6, 1)	(3, 4, 5, 7, 0)	0.30064206
(2, 7, 0)	(3, 4, 5, 6, 1)	0.31145588
(2, 7, 1)	(3, 4, 5, 6, 0)	0.31145588
(2, 0, 1)	(3, 4, 5, 6, 7)	0.42756594
(2, 3, 4, 5)	(6, 7, 0, 1)	0.47865643
(2, 3, 4, 6)	(5, 7, 0, 1)	0.27204245
(2, 3, 4, 7)	(5, 6, 0, 1)	0.29330035
(2, 3, 4, 0)	(5, 6, 7, 1)	0.27204245
(2, 3, 4, 1)	(5, 6, 7, 0)	0.47865643
(2, 3, 5, 6)	(4, 7, 0, 1)	0.29330035
(2, 3, 5, 7)	(4, 6, 0, 1)	0.20203136
(2, 3, 5, 0)	(4, 6, 7, 1)	0.20033735
(2, 3, 5, 1)	(4, 6, 7, 0)	0.27204245
(2, 3, 6, 7)	(4, 5, 0, 1)	0.35325200
(2, 3, 6, 0)	(4, 5, 7, 1)	0.20203136
(2, 3, 6, 1)	(4, 5, 7, 0)	0.29330035
(2, 3, 7, 0)	(4, 5, 6, 1)	0.29330035
(2, 3, 7, 1)	(4, 5, 6, 0)	0.27204245
(2, 3, 0, 1)	(4, 5, 6, 7)	0.47865643
(2, 4, 5, 6)	(3, 7, 0, 1)	0.27204245
(2, 4, 5, 7)	(3, 6, 0, 1)	0.20033735
(2, 4, 5, 0)	(3, 6, 7, 1)	0.20203136
(2, 4, 5, 1)	(3, 6, 7, 0)	0.29330035
(2, 4, 6, 7)	(3, 5, 0, 1)	0.20203136
(2, 4, 6, 0)	(3, 5, 7, 1)	0.13053255
(2, 4, 6, 1)	(3, 5, 7, 0)	0.20203136
(2, 4, 7, 0)	(3, 5, 6, 1)	0.20203136
(2, 4, 7, 1)	(3, 5, 6, 0)	0.20033735
(2, 4, 0, 1)	(3, 5, 6, 7)	0.27204245
(2, 5, 6, 7)	(3, 4, 0, 1)	0.29330035
(2, 5, 6, 0)	(3, 4, 7, 1)	0.20203136
(2, 5, 6, 1)	(3, 4, 7, 0)	0.35325200
(2, 5, 7, 0)	(3, 4, 6, 1)	0.20033735
(2, 5, 7, 1)	(3, 4, 6, 0)	0.20203136
(2, 5, 0, 1)	(3, 4, 6, 7)	0.29330035
(2, 6, 7, 0)	(3, 4, 5, 1)	0.27204245
(2, 6, 7, 1)	(3, 4, 5, 0)	0.29330035
(2, 6, 0, 1)	(3, 4, 5, 7)	0.27204245
(2, 7, 0, 1)	(3, 4, 5, 6)	0.47865643
(2, 3, 4, 5, 6)	(7, 0, 1)	0.42756594
(2, 3, 4, 5, 7)	(6, 0, 1)	0.31145588
(2, 3, 4, 5, 0)	(6, 7, 1)	0.31145588
(2, 3, 4, 5, 1)	(6, 7, 0)	0.42756594
(2, 3, 4, 6, 7)	(5, 0, 1)	0.30064206
(2, 3, 4, 6, 0)	(5, 7, 1)	0.18889328
(2, 3, 4, 6, 1)	(5, 7, 0)	0.31145588
(2, 3, 4, 7, 0)	(5, 6, 1)	0.30064206
(2, 3, 4, 7, 1)	(5, 6, 0)	0.31145588
(2, 3, 4, 0, 1)	(5, 6, 7)	0.42756594
(2, 3, 5, 6, 7)	(4, 0, 1)	0.30064206
(2, 3, 5, 6, 0)	(4, 7, 1)	0.19534582
(2, 3, 5, 6, 1)	(4, 7, 0)	0.30064206
(2, 3, 5, 7, 0)	(4, 6, 1)	0.19534582
(2, 3, 5, 7, 1)	(4, 6, 0)	0.18889328
(2, 3, 5, 0, 1)	(4, 6, 7)	0.31145588
(2, 3, 6, 7, 0)	(4, 5, 1)	0.30064206
(2, 3, 6, 7, 1)	(4, 5, 0)	0.30064206
(2, 3, 6, 0, 1)	(4, 5, 7)	0.31145588
(2, 3, 7, 0, 1)	(4, 5, 6)	0.42756594
(2, 4, 5, 6, 7)	(3, 0, 1)	0.31145588
(2, 4, 5, 6, 0)	(3, 7, 1)	0.18889328
(2, 4, 5, 6, 1)	(3, 7, 0)	0.30064206
(2, 4, 5, 7, 0)	(3, 6, 1)	0.19534582
(2, 4, 5, 7, 1)	(3, 6, 0)	0.19534582
(2, 4, 5, 0, 1)	(3, 6, 7)	0.30064206
(2, 4, 6, 7, 0)	(3, 5, 1)	0.18889328
(2, 4, 6, 7, 1)	(3, 5, 0)	0.19534582
(2, 4, 6, 0, 1)	(3, 5, 7)	0.18889328
(2, 4, 7, 0, 1)	(3, 5, 6)	0.31145588
(2, 5, 6, 7, 0)	(3, 4, 1)	0.31145588
(2, 5, 6, 7, 1)	(3, 4, 0)	0.30064206
(2, 5, 6, 0, 1)	(3, 4, 7)	0.30064206
(2, 5, 7, 0, 1)	(3, 4, 6)	0.31145588
(2, 6, 7, 0, 1)	(3, 4, 5)	0.42756594
(2, 3, 4, 5, 6, 7)	(0, 1)	0.52766357
(2, 3, 4, 5, 6, 0)	(7, 1)	0.29980473
(2, 3, 4, 5, 6, 1)	(7, 0)	0.52766357
(2, 3, 4, 5, 7, 0)	(6, 1)	0.29544345
(2, 3, 4, 5, 7, 1)	(6, 0)	0.29980473
(2, 3, 4, 5, 0, 1)	(6, 7)	0.52766357
(2, 3, 4, 6, 7, 0)	(5, 1)	0.27817710
(2, 3, 4, 6, 7, 1)	(5, 0)	0.29544345
(2, 3, 4, 6, 0, 1)	(5, 7)	0.29980473
(2, 3, 4, 7, 0, 1)	(5, 6)	0.52766357
(2, 3, 5, 6, 7, 0)	(4, 1)	0.29544345
(2, 3, 5, 6, 7, 1)	(4, 0)	0.27817710
(2, 3, 5, 6, 0, 1)	(4, 7)	0.29544345
(2, 3, 5, 7, 0, 1)	(4, 6)	0.29980473
(2, 3, 6, 7, 0, 1)	(4, 5)	0.52766357
(2, 4, 5, 6, 7, 0)	(3, 1)	0.29980473
(2, 4, 5, 6, 7, 1)	(3, 0)	0.29544345
(2, 4, 5, 6, 0, 1)	(3, 7)	0.27817710
(2, 4, 5, 7, 0, 1)	(3, 6)	0.29544345
(2, 4, 6, 7, 0, 1)	(3, 5)	0.29980473
(2, 5, 6, 7, 0, 1)	(3, 4)	0.52766357
(2, 3, 4, 5, 6, 7, 0)	(1,)	0.50000000
(2, 3, 4, 5, 6, 7, 1)	(0,)	0.50000000
(2, 3, 4, 5, 6, 0, 1)	(7,)	0.50000000
(2, 3, 4, 5, 7, 0, 1)	(6,)	0.50000000
(2, 3, 4, 6, 7, 0, 1)	(5,)	0.50000000
(2, 3, 5, 6, 7, 0, 1)	(4,)	0.50000000
(2, 4, 5, 6, 7, 0, 1)	(3,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.97194283
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

- C_exact = 0.97194283
- C_LB = 0.31394240
- Gap = 0.65800042


8-Body RDM (Sites 3,4,5,6,7,0,1,2)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 8-Body RDM (Sites 3,4,5,6,7,0,1,2)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000001
Eigenvalue min	-2.661793e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

Convex Roof Analysis
==========

Lower Bounds:
- Paper (witness) = 0.00000000
- Notes (6-9) = 0.31394240
- Best C_LB = 0.31394240

C_GME (unitary optimization - Cayley Transform):
- C_num (unitary opt) = 0.97194283 BEST
- (20 restarts)

C_GME (unitary optimization - Expoenetial Map):
- C_num = 0.97194283 BEST
- (L-BFGS-B, 20 restarts)

Upper Bound:
- C_UB (eigendecomp) = 0.97194283

Status: GME
Bounds consistent: YES
Convergence: TIGHT

GME Analysis
==========

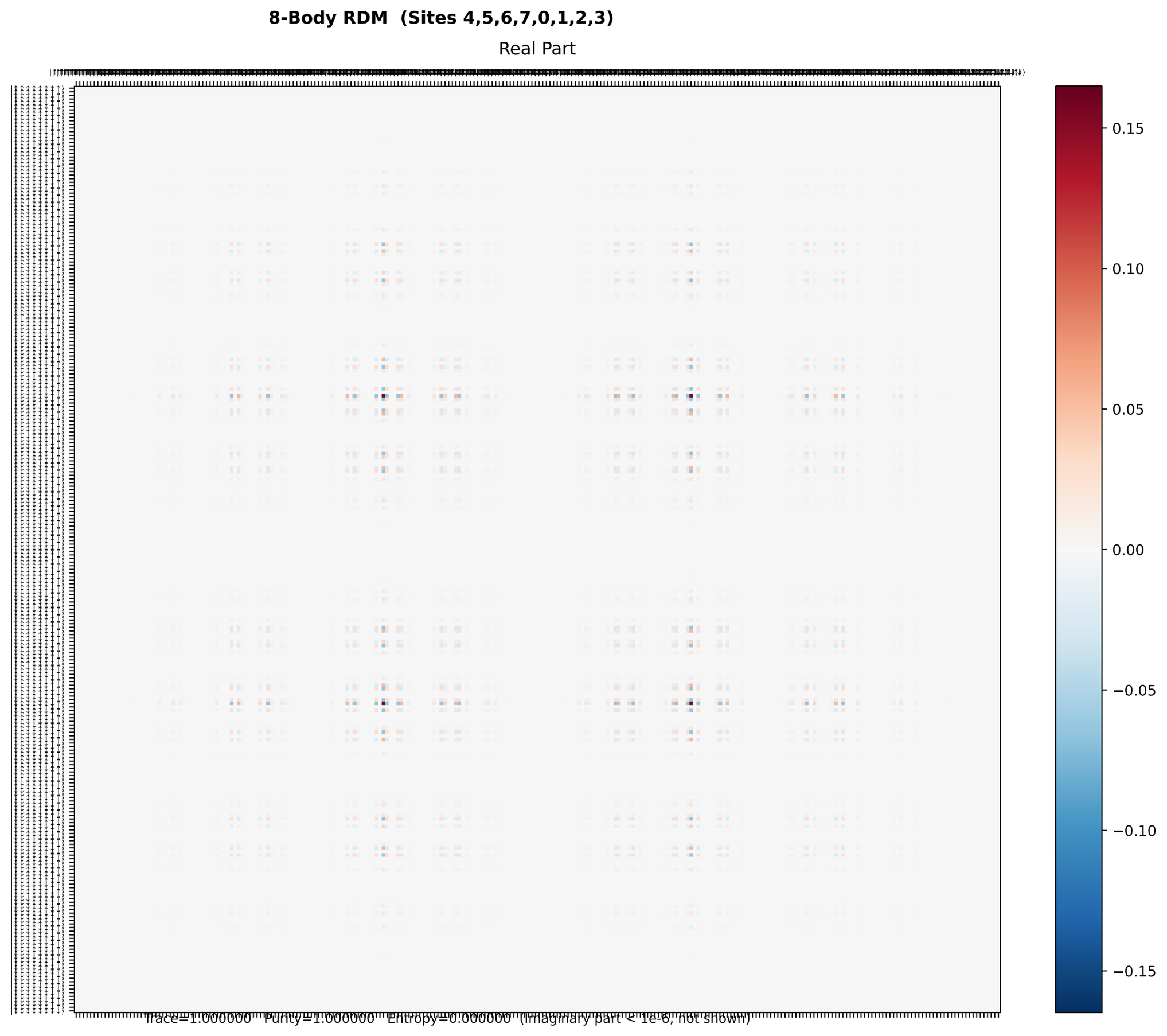
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(3,)	(4, 5, 6, 7, 0, 1, 2)	0.50000000
(3, 4)	(5, 6, 7, 0, 1, 2)	0.52766357
(3, 5)	(4, 6, 7, 0, 1, 2)	0.29980473
(3, 6)	(4, 5, 7, 0, 1, 2)	0.29544345
(3, 7)	(4, 5, 6, 0, 1, 2)	0.27817710
(3, 0)	(4, 5, 6, 7, 1, 2)	0.29544345
(3, 1)	(4, 5, 6, 7, 0, 2)	0.29980473
(3, 2)	(4, 5, 6, 7, 0, 1)	0.52766357
(3, 4, 5)	(6, 7, 0, 1, 2)	0.42756594
(3, 4, 6)	(5, 7, 0, 1, 2)	0.31145588
(3, 4, 7)	(5, 6, 0, 1, 2)	0.30064206
(3, 4, 0)	(5, 6, 7, 1, 2)	0.30064206
(3, 4, 1)	(5, 6, 7, 0, 2)	0.31145588
(3, 4, 2)	(5, 6, 7, 0, 1)	0.42756594
(3, 5, 6)	(4, 7, 0, 1, 2)	0.31145588
(3, 5, 7)	(4, 6, 0, 1, 2)	0.18889328
(3, 5, 0)	(4, 6, 7, 1, 2)	0.19534582
(3, 5, 1)	(4, 6, 7, 0, 2)	0.18889328
(3, 5, 2)	(4, 6, 7, 0, 1)	0.31145588
(3, 6, 7)	(4, 5, 0, 1, 2)	0.30064206
(3, 6, 0)	(4, 5, 7, 1, 2)	0.19534582
(3, 6, 1)	(4, 5, 7, 0, 2)	0.19534582
(3, 6, 2)	(4, 5, 7, 0, 1)	0.30064206
(3, 7, 0)	(4, 5, 6, 1, 2)	0.30064206
(3, 7, 1)	(4, 5, 6, 0, 2)	0.18889328
(3, 7, 2)	(4, 5, 6, 0, 1)	0.30064206
(3, 0, 1)	(4, 5, 6, 7, 2)	0.31145588
(3, 0, 2)	(4, 5, 6, 7, 1)	0.31145588
(3, 1, 2)	(4, 5, 6, 7, 0)	0.42756594
(3, 4, 5, 6)	(7, 0, 1, 2)	0.47865643
(3, 4, 5, 7)	(6, 0, 1, 2)	0.27204245
(3, 4, 5, 0)	(6, 7, 1, 2)	0.29330035
(3, 4, 5, 1)	(6, 7, 0, 2)	0.27204245
(3, 4, 5, 2)	(6, 7, 0, 1)	0.47865643
(3, 4, 6, 7)	(5, 0, 1, 2)	0.29330035
(3, 4, 6, 0)	(5, 7, 1, 2)	0.20203136
(3, 4, 6, 1)	(5, 7, 0, 2)	0.20033735
(3, 4, 6, 2)	(5, 7, 0, 1)	0.27204245
(3, 4, 7, 0)	(5, 6, 1, 2)	0.35325200
(3, 4, 7, 1)	(5, 6, 0, 2)	0.20203136
(3, 4, 7, 2)	(5, 6, 0, 1)	0.29330035
(3, 4, 0, 1)	(5, 6, 7, 2)	0.29330035
(3, 4, 0, 2)	(5, 6, 7, 1)	0.27204245
(3, 4, 1, 2)	(5, 6, 7, 0)	0.47865643
(3, 5, 6, 7)	(4, 0, 1, 2)	0.27204245
(3, 5, 6, 0)	(4, 7, 1, 2)	0.20033735
(3, 5, 6, 1)	(4, 7, 0, 2)	0.20203136
(3, 5, 6, 2)	(4, 7, 0, 1)	0.29330035
(3, 5, 7, 0)	(4, 6, 1, 2)	0.20203136
(3, 5, 7, 1)	(4, 6, 0, 2)	0.13053255
(3, 5, 7, 2)	(4, 6, 0, 1)	0.20203136
(3, 5, 0, 1)	(4, 6, 7, 2)	0.20203136
(3, 5, 0, 2)	(4, 6, 7, 1)	0.20033735
(3, 5, 1, 2)	(4, 6, 7, 0)	0.27204245
(3, 6, 7, 0)	(4, 5, 1, 2)	0.29330035
(3, 6, 7, 1)	(4, 5, 0, 2)	0.20203136
(3, 6, 7, 2)	(4, 5, 0, 1)	0.35325200
(3, 6, 0, 1)	(4, 5, 7, 2)	0.20033735
(3, 6, 0, 2)	(4, 5, 7, 1)	0.20203136
(3, 6, 1, 2)	(4, 5, 7, 0)	0.29330035
(3, 7, 0, 1)	(4, 5, 6, 2)	0.27204245
(3, 7, 0, 2)	(4, 5, 6, 1)	0.29330035
(3, 7, 1, 2)	(4, 5, 6, 0)	0.27204245
(3, 0, 1, 2)	(4, 5, 6, 7)	0.47865643
(3, 4, 5, 6, 7)	(0, 1, 2)	0.42756594
(3, 4, 5, 6, 0)	(7, 1, 2)	0.31145588
(3, 4, 5, 6, 1)	(7, 0, 2)	0.31145588
(3, 4, 5, 6, 2)	(7, 0, 1)	0.42756594
(3, 4, 5, 7, 0)	(6, 1, 2)	0.30064206
(3, 4, 5, 7, 1)	(6, 0, 2)	0.18889328
(3, 4, 5, 7, 2)	(6, 0, 1)	0.31145588
(3, 4, 5, 0, 1)	(6, 7, 2)	0.30064206
(3, 4, 5, 0, 2)	(6, 7, 1)	0.31145588
(3, 4, 5, 1, 2)	(6, 7, 0)	0.42756594
(3, 4, 6, 7, 0)	(5, 1, 2)	0.30064206
(3, 4, 6, 7, 1)	(5, 0, 2)	0.19534582
(3, 4, 6, 7, 2)	(5, 0, 1)	0.30064206
(3, 4, 6, 0, 1)	(5, 7, 2)	0.19534582
(3, 4, 6, 0, 2)	(5, 7, 1)	0.18889328
(3, 4, 6, 1, 2)	(5, 7, 0)	0.31145588
(3, 4, 7, 0, 1)	(5, 6, 2)	0.30064206
(3, 4, 7, 0, 2)	(5, 6, 1)	0.30064206
(3, 4, 7, 1, 2)	(5, 6, 0)	0.31145588
(3, 4, 0, 1, 2)	(5, 6, 7)	0.42756594
(3, 5, 6, 7, 0)	(4, 1, 2)	0.31145588
(3, 5, 6, 7, 1)	(4, 0, 2)	0.18889328
(3, 5, 6, 7, 2)	(4, 0, 1)	0.30064206
(3, 5, 6, 0, 1)	(4, 7, 2)	0.19534582
(3, 5, 6, 0, 2)	(4, 7, 1)	0.19534582
(3, 5, 6, 1, 2)	(4, 7, 0)	0.30064206
(3, 5, 7, 0, 1)	(4, 6, 2)	0.18889328
(3, 5, 7, 0, 2)	(4, 6, 1)	0.19534582
(3, 5, 7, 1, 2)	(4, 6, 0)	0.18889328
(3, 5, 0, 1, 2)	(4, 6, 7)	0.31145588
(3, 6, 7, 0, 1)	(4, 5, 2)	0.31145588
(3, 6, 7, 0, 2)	(4, 5, 1)	0.30064206
(3, 6, 7, 1, 2)	(4, 5, 0)	0.30064206
(3, 6, 0, 1, 2)	(4, 5, 7)	0.31145588
(3, 7, 0, 1, 2)	(4, 5, 6)	0.42756594
(3, 4, 5, 6, 7, 0)	(1, 2)	0.52766357
(3, 4, 5, 6, 7, 1)	(0, 2)	0.29980473
(3, 4, 5, 6, 7, 2)	(0, 1)	0.52766357
(3, 4, 5, 6, 0, 1)	(7, 2)	0.29544345
(3, 4, 5, 6, 0, 2)	(7, 1)	0.29980473
(3, 4, 5, 6, 1, 2)	(7, 0)	0.52766357
(3, 4, 5, 7, 0, 1)	(6, 2)	0.27817710
(3, 4, 5, 7, 0, 2)	(6, 1)	0.29544345
(3, 4, 5, 7, 1, 2)	(6, 0)	0.29980473
(3, 4, 5, 0, 1, 2)	(6, 7)	0.52766357
(3, 4, 6, 7, 0, 1)	(5, 2)	0.29544345
(3, 4, 6, 7, 0, 2)	(5, 1)	0.27817710
(3, 4, 6, 7, 1, 2)	(5, 0)	0.29544345
(3, 4, 6, 0, 1, 2)	(5, 7)	0.29980473
(3, 4, 7, 0, 1, 2)	(5, 6)	0.52766357
(3, 5, 6, 7, 0, 1)	(4, 2)	0.29980473
(3, 5, 6, 7, 0, 2)	(4, 1)	0.29544345
(3, 5, 6, 7, 1, 2)	(4, 0)	0.27817710
(3, 5, 6, 0, 1, 2)	(4, 7)	0.29544345
(3, 5, 7, 0, 1, 2)	(4, 6)	0.29980473
(3, 6, 7, 0, 1, 2)	(4, 5)	0.52766357
(3, 4, 5, 6, 7, 0, 1)	(2,)	0.50000000
(3, 4, 5, 6, 7, 0, 2)	(1,)	0.50000000
(3, 4, 5, 6, 7, 1, 2)	(0,)	0.50000000
(3, 4, 5, 6, 0, 1, 2)	(7,)	0.50000000
(3, 4, 5, 7, 0, 1, 2)	(6,)	0.50000000
(3, 4, 6, 7, 0, 1, 2)	(5,)	0.50000000
(3, 5, 6, 7, 0, 1, 2)	(4,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.97194283
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

- C_exact = 0.97194283
- C_LB = 0.31394240
- Gap = 0.65800042


8-Body RDM (Sites 4,5,6,7,0,1,2,3)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 8-Body RDM (Sites 4,5,6,7,0,1,2,3)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000001
Eigenvalue min	-3.465744e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

Convex Roof Analysis
==========

Lower Bounds:
- Paper (witness) = 0.00000000
- Notes (6-9) = 0.31394240
- Best C_LB = 0.31394240

C_GME (unitary optimization - Cayley Transform):
- C_num (unitary opt) = 0.97194283 BEST
- (20 restarts)

C_GME (unitary optimization - Expoenetial Map):
- C_num = 0.97194283 BEST
- (L-BFGS-B, 20 restarts)

Upper Bound:
- C_UB (eigendecomp) = 0.97194283

Status: GME
Bounds consistent: YES
Convergence: TIGHT

GME Analysis
==========

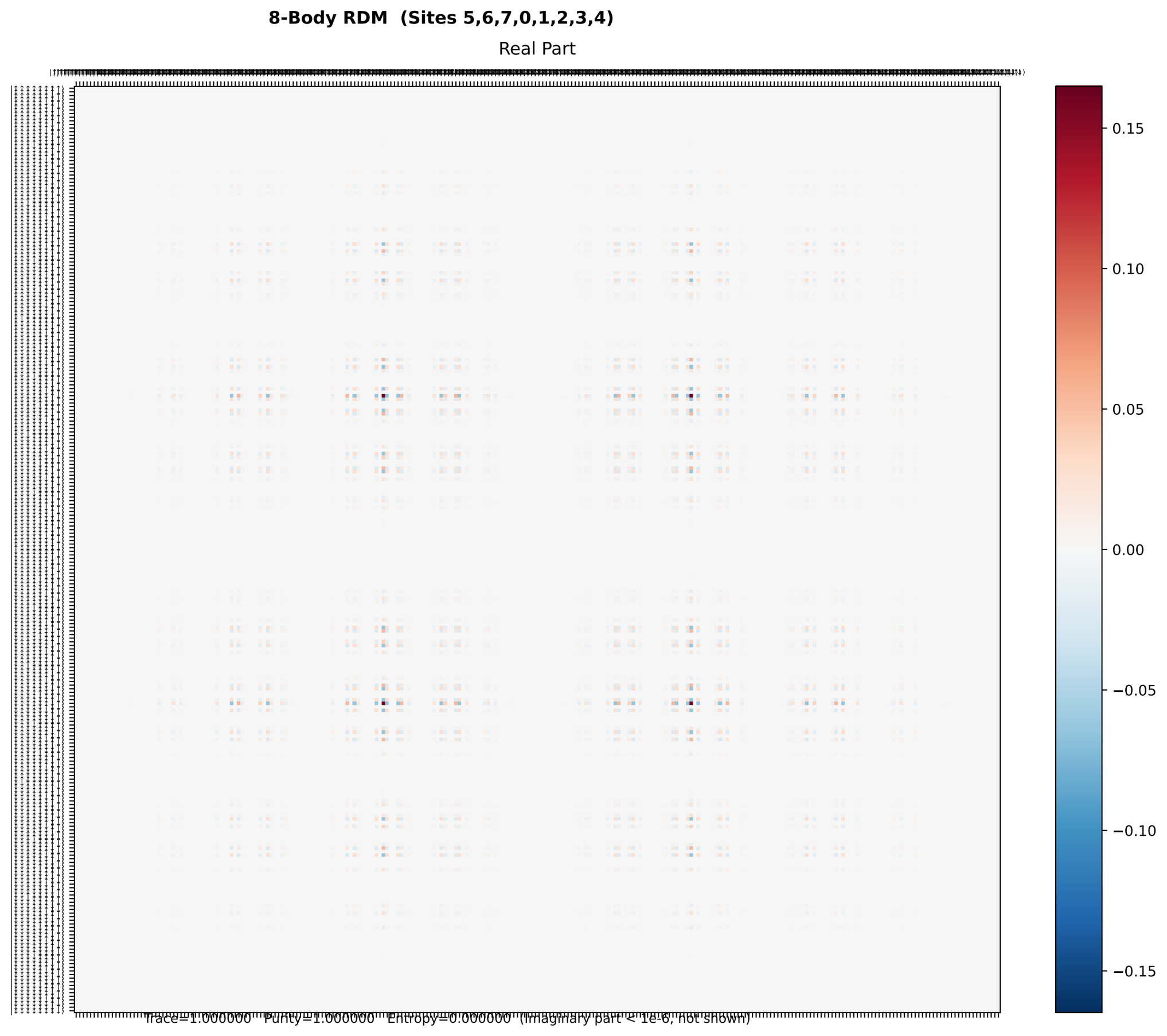
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(4,)	(5, 6, 7, 0, 1, 2, 3)	0.50000000
(4, 5)	(6, 7, 0, 1, 2, 3)	0.52766357
(4, 6)	(5, 7, 0, 1, 2, 3)	0.29980473
(4, 7)	(5, 6, 0, 1, 2, 3)	0.29544345
(4, 0)	(5, 6, 7, 1, 2, 3)	0.27817710
(4, 1)	(5, 6, 7, 0, 2, 3)	0.29544345
(4, 2)	(5, 6, 7, 0, 1, 3)	0.29980473
(4, 3)	(5, 6, 7, 0, 1, 2)	0.52766357
(4, 5, 6)	(7, 0, 1, 2, 3)	0.42756594
(4, 5, 7)	(6, 0, 1, 2, 3)	0.31145588
(4, 5, 0)	(6, 7, 1, 2, 3)	0.30064206
(4, 5, 1)	(6, 7, 0, 2, 3)	0.30064206
(4, 5, 2)	(6, 7, 0, 1, 3)	0.31145588
(4, 5, 3)	(6, 7, 0, 1, 2)	0.42756594
(4, 6, 7)	(5, 0, 1, 2, 3)	0.31145588
(4, 6, 0)	(5, 7, 1, 2, 3)	0.18889328
(4, 6, 1)	(5, 7, 0, 2, 3)	0.19534582
(4, 6, 2)	(5, 7, 0, 1, 3)	0.18889328
(4, 6, 3)	(5, 7, 0, 1, 2)	0.31145588
(4, 7, 0)	(5, 6, 1, 2, 3)	0.30064206
(4, 7, 1)	(5, 6, 0, 2, 3)	0.19534582
(4, 7, 2)	(5, 6, 0, 1, 3)	0.19534582
(4, 7, 3)	(5, 6, 0, 1, 2)	0.30064206
(4, 0, 1)	(5, 6, 7, 2, 3)	0.30064206
(4, 0, 2)	(5, 6, 7, 1, 3)	0.18889328
(4, 0, 3)	(5, 6, 7, 1, 2)	0.30064206
(4, 1, 2)	(5, 6, 7, 0, 3)	0.31145588
(4, 1, 3)	(5, 6, 7, 0, 2)	0.31145588
(4, 2, 3)	(5, 6, 7, 0, 1)	0.42756594
(4, 5, 6, 7)	(0, 1, 2, 3)	0.47865643
(4, 5, 6, 0)	(7, 1, 2, 3)	0.27204245
(4, 5, 6, 1)	(7, 0, 2, 3)	0.29330035
(4, 5, 6, 2)	(7, 0, 1, 3)	0.27204245
(4, 5, 6, 3)	(7, 0, 1, 2)	0.47865643
(4, 5, 7, 0)	(6, 1, 2, 3)	0.29330035
(4, 5, 7, 1)	(6, 0, 2, 3)	0.20203136
(4, 5, 7, 2)	(6, 0, 1, 3)	0.20033735
(4, 5, 7, 3)	(6, 0, 1, 2)	0.27204245
(4, 5, 0, 1)	(6, 7, 2, 3)	0.35325200
(4, 5, 0, 2)	(6, 7, 1, 3)	0.20203136
(4, 5, 0, 3)	(6, 7, 1, 2)	0.29330035
(4, 5, 1, 2)	(6, 7, 0, 3)	0.29330035
(4, 5, 1, 3)	(6, 7, 0, 2)	0.27204245
(4, 5, 2, 3)	(6, 7, 0, 1)	0.47865643
(4, 6, 7, 0)	(5, 1, 2, 3)	0.27204245
(4, 6, 7, 1)	(5, 0, 2, 3)	0.20033735
(4, 6, 7, 2)	(5, 0, 1, 3)	0.20203136
(4, 6, 7, 3)	(5, 0, 1, 2)	0.29330035
(4, 6, 0, 1)	(5, 7, 2, 3)	0.20203136
(4, 6, 0, 2)	(5, 7, 1, 3)	0.13053255
(4, 6, 0, 3)	(5, 7, 1, 2)	0.20203136
(4, 6, 1, 2)	(5, 7, 0, 3)	0.20203136
(4, 6, 1, 3)	(5, 7, 0, 2)	0.20033735
(4, 6, 2, 3)	(5, 7, 0, 1)	0.27204245
(4, 7, 0, 1)	(5, 6, 2, 3)	0.29330035
(4, 7, 0, 2)	(5, 6, 1, 3)	0.20203136
(4, 7, 0, 3)	(5, 6, 1, 2)	0.35325200
(4, 7, 1, 2)	(5, 6, 0, 3)	0.20033735
(4, 7, 1, 3)	(5, 6, 0, 2)	0.20203136
(4, 7, 2, 3)	(5, 6, 0, 1)	0.29330035
(4, 0, 1, 2)	(5, 6, 7, 3)	0.27204245
(4, 0, 1, 3)	(5, 6, 7, 2)	0.29330035
(4, 0, 2, 3)	(5, 6, 7, 1)	0.27204245
(4, 1, 2, 3)	(5, 6, 7, 0)	0.47865643
(4, 5, 6, 7, 0)	(1, 2, 3)	0.42756594
(4, 5, 6, 7, 1)	(0, 2, 3)	0.31145588
(4, 5, 6, 7, 2)	(0, 1, 3)	0.31145588
(4, 5, 6, 7, 3)	(0, 1, 2)	0.42756594
(4, 5, 6, 0, 1)	(7, 2, 3)	0.30064206
(4, 5, 6, 0, 2)	(7, 1, 3)	0.18889328
(4, 5, 6, 0, 3)	(7, 1, 2)	0.31145588
(4, 5, 6, 1, 2)	(7, 0, 3)	0.30064206
(4, 5, 6, 1, 3)	(7, 0, 2)	0.31145588
(4, 5, 6, 2, 3)	(7, 0, 1)	0.42756594
(4, 5, 7, 0, 1)	(6, 2, 3)	0.30064206
(4, 5, 7, 0, 2)	(6, 1, 3)	0.19534582
(4, 5, 7, 0, 3)	(6, 1, 2)	0.30064206
(4, 5, 7, 1, 2)	(6, 0, 3)	0.19534582
(4, 5, 7, 1, 3)	(6, 0, 2)	0.18889328
(4, 5, 7, 2, 3)	(6, 0, 1)	0.31145588
(4, 5, 0, 1, 2)	(6, 7, 3)	0.30064206
(4, 5, 0, 1, 3)	(6, 7, 2)	0.30064206
(4, 5, 0, 2, 3)	(6, 7, 1)	0.31145588
(4, 5, 1, 2, 3)	(6, 7, 0)	0.42756594
(4, 6, 7, 0, 1)	(5, 2, 3)	0.31145588
(4, 6, 7, 0, 2)	(5, 1, 3)	0.18889328
(4, 6, 7, 0, 3)	(5, 1, 2)	0.30064206
(4, 6, 7, 1, 2)	(5, 0, 3)	0.19534582
(4, 6, 7, 1, 3)	(5, 0, 2)	0.19534582
(4, 6, 7, 2, 3)	(5, 0, 1)	0.30064206
(4, 6, 0, 1, 2)	(5, 7, 3)	0.18889328
(4, 6, 0, 1, 3)	(5, 7, 2)	0.19534582
(4, 6, 0, 2, 3)	(5, 7, 1)	0.18889328
(4, 6, 1, 2, 3)	(5, 7, 0)	0.31145588
(4, 7, 0, 1, 2)	(5, 6, 3)	0.31145588
(4, 7, 0, 1, 3)	(5, 6, 2)	0.30064206
(4, 7, 0, 2, 3)	(5, 6, 1)	0.30064206
(4, 7, 1, 2, 3)	(5, 6, 0)	0.31145588
(4, 0, 1, 2, 3)	(5, 6, 7)	0.42756594
(4, 5, 6, 7, 0, 1)	(2, 3)	0.52766357
(4, 5, 6, 7, 0, 2)	(1, 3)	0.29980473
(4, 5, 6, 7, 0, 3)	(1, 2)	0.52766357
(4, 5, 6, 7, 1, 2)	(0, 3)	0.29544345
(4, 5, 6, 7, 1, 3)	(0, 2)	0.29980473
(4, 5, 6, 7, 2, 3)	(0, 1)	0.52766357
(4, 5, 6, 0, 1, 2)	(7, 3)	0.27817710
(4, 5, 6, 0, 1, 3)	(7, 2)	0.29544345
(4, 5, 6, 0, 2, 3)	(7, 1)	0.29980473
(4, 5, 6, 1, 2, 3)	(7, 0)	0.52766357
(4, 5, 7, 0, 1, 2)	(6, 3)	0.29544345
(4, 5, 7, 0, 1, 3)	(6, 2)	0.27817710
(4, 5, 7, 0, 2, 3)	(6, 1)	0.29544345
(4, 5, 7, 1, 2, 3)	(6, 0)	0.29980473
(4, 5, 0, 1, 2, 3)	(6, 7)	0.52766357
(4, 6, 7, 0, 1, 2)	(5, 3)	0.29980473
(4, 6, 7, 0, 1, 3)	(5, 2)	0.29544345
(4, 6, 7, 0, 2, 3)	(5, 1)	0.27817710
(4, 6, 7, 1, 2, 3)	(5, 0)	0.29544345
(4, 6, 0, 1, 2, 3)	(5, 7)	0.29980473
(4, 7, 0, 1, 2, 3)	(5, 6)	0.52766357
(4, 5, 6, 7, 0, 1, 2)	(3,)	0.50000000
(4, 5, 6, 7, 0, 1, 3)	(2,)	0.50000000
(4, 5, 6, 7, 0, 2, 3)	(1,)	0.50000000
(4, 5, 6, 7, 1, 2, 3)	(0,)	0.50000000
(4, 5, 6, 0, 1, 2, 3)	(7,)	0.50000000
(4, 5, 7, 0, 1, 2, 3)	(6,)	0.50000000
(4, 6, 7, 0, 1, 2, 3)	(5,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.97194283
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

- C_exact = 0.97194283
- C_LB = 0.31394240
- Gap = 0.65800042


8-Body RDM (Sites 5,6,7,0,1,2,3,4)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 8-Body RDM (Sites 5,6,7,0,1,2,3,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000001
Eigenvalue min	-3.296811e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

Convex Roof Analysis
==========

Lower Bounds:
- Paper (witness) = 0.00000000
- Notes (6-9) = 0.31394240
- Best C_LB = 0.31394240

C_GME (unitary optimization - Cayley Transform):
- C_num (unitary opt) = 0.97194283 BEST
- (20 restarts)

C_GME (unitary optimization - Expoenetial Map):
- C_num = 0.97194283 BEST
- (L-BFGS-B, 20 restarts)

Upper Bound:
- C_UB (eigendecomp) = 0.97194283

Status: GME
Bounds consistent: YES
Convergence: TIGHT

GME Analysis
==========

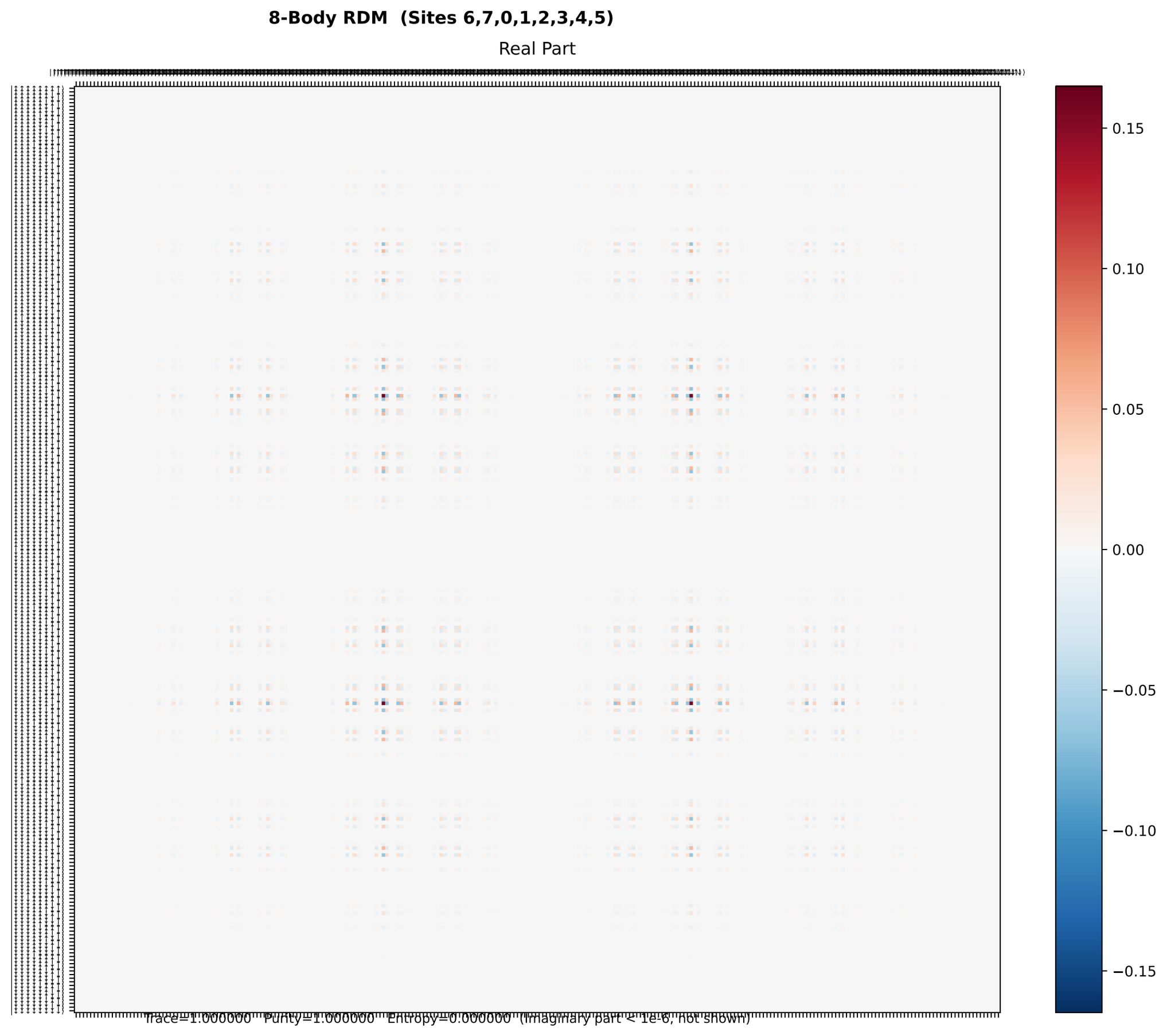
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(5,)	(6, 7, 0, 1, 2, 3, 4)	0.50000000
(5, 6)	(7, 0, 1, 2, 3, 4)	0.52766357
(5, 7)	(6, 0, 1, 2, 3, 4)	0.29980473
(5, 0)	(6, 7, 1, 2, 3, 4)	0.29544345
(5, 1)	(6, 7, 0, 2, 3, 4)	0.27817710
(5, 2)	(6, 7, 0, 1, 3, 4)	0.29544345
(5, 3)	(6, 7, 0, 1, 2, 4)	0.29980473
(5, 4)	(6, 7, 0, 1, 2, 3)	0.52766357
(5, 6, 7)	(0, 1, 2, 3, 4)	0.42756594
(5, 6, 0)	(7, 1, 2, 3, 4)	0.31145588
(5, 6, 1)	(7, 0, 2, 3, 4)	0.30064206
(5, 6, 2)	(7, 0, 1, 3, 4)	0.30064206
(5, 6, 3)	(7, 0, 1, 2, 4)	0.31145588
(5, 6, 4)	(7, 0, 1, 2, 3)	0.42756594
(5, 7, 0)	(6, 1, 2, 3, 4)	0.31145588
(5, 7, 1)	(6, 0, 2, 3, 4)	0.18889328
(5, 7, 2)	(6, 0, 1, 3, 4)	0.19534582
(5, 7, 3)	(6, 0, 1, 2, 4)	0.18889328
(5, 7, 4)	(6, 0, 1, 2, 3)	0.31145588
(5, 0, 1)	(6, 7, 2, 3, 4)	0.30064206
(5, 0, 2)	(6, 7, 1, 3, 4)	0.19534582
(5, 0, 3)	(6, 7, 1, 2, 4)	0.19534582
(5, 0, 4)	(6, 7, 1, 2, 3)	0.30064206
(5, 1, 2)	(6, 7, 0, 3, 4)	0.30064206
(5, 1, 3)	(6, 7, 0, 2, 4)	0.18889328
(5, 1, 4)	(6, 7, 0, 2, 3)	0.30064206
(5, 2, 3)	(6, 7, 0, 1, 4)	0.31145588
(5, 2, 4)	(6, 7, 0, 1, 3)	0.31145588
(5, 3, 4)	(6, 7, 0, 1, 2)	0.42756594
(5, 6, 7, 0)	(1, 2, 3, 4)	0.47865643
(5, 6, 7, 1)	(0, 2, 3, 4)	0.27204245
(5, 6, 7, 2)	(0, 1, 3, 4)	0.29330035
(5, 6, 7, 3)	(0, 1, 2, 4)	0.27204245
(5, 6, 7, 4)	(0, 1, 2, 3)	0.47865643
(5, 6, 0, 1)	(7, 2, 3, 4)	0.29330035
(5, 6, 0, 2)	(7, 1, 3, 4)	0.20203136
(5, 6, 0, 3)	(7, 1, 2, 4)	0.20033735
(5, 6, 0, 4)	(7, 1, 2, 3)	0.27204245
(5, 6, 1, 2)	(7, 0, 3, 4)	0.35325200
(5, 6, 1, 3)	(7, 0, 2, 4)	0.20203136
(5, 6, 1, 4)	(7, 0, 2, 3)	0.29330035
(5, 6, 2, 3)	(7, 0, 1, 4)	0.29330035
(5, 6, 2, 4)	(7, 0, 1, 3)	0.27204245
(5, 6, 3, 4)	(7, 0, 1, 2)	0.47865643
(5, 7, 0, 1)	(6, 2, 3, 4)	0.27204245
(5, 7, 0, 2)	(6, 1, 3, 4)	0.20033735
(5, 7, 0, 3)	(6, 1, 2, 4)	0.20203136
(5, 7, 0, 4)	(6, 1, 2, 3)	0.29330035
(5, 7, 1, 2)	(6, 0, 3, 4)	0.20203136
(5, 7, 1, 3)	(6, 0, 2, 4)	0.13053255
(5, 7, 1, 4)	(6, 0, 2, 3)	0.20203136
(5, 7, 2, 3)	(6, 0, 1, 4)	0.20203136
(5, 7, 2, 4)	(6, 0, 1, 3)	0.20033735
(5, 7, 3, 4)	(6, 0, 1, 2)	0.27204245
(5, 0, 1, 2)	(6, 7, 3, 4)	0.29330035
(5, 0, 1, 3)	(6, 7, 2, 4)	0.20203136
(5, 0, 1, 4)	(6, 7, 2, 3)	0.35325200
(5, 0, 2, 3)	(6, 7, 1, 4)	0.20033735
(5, 0, 2, 4)	(6, 7, 1, 3)	0.20203136
(5, 0, 3, 4)	(6, 7, 1, 2)	0.29330035
(5, 1, 2, 3)	(6, 7, 0, 4)	0.27204245
(5, 1, 2, 4)	(6, 7, 0, 3)	0.29330035
(5, 1, 3, 4)	(6, 7, 0, 2)	0.27204245
(5, 2, 3, 4)	(6, 7, 0, 1)	0.47865643
(5, 6, 7, 0, 1)	(2, 3, 4)	0.42756594
(5, 6, 7, 0, 2)	(1, 3, 4)	0.31145588
(5, 6, 7, 0, 3)	(1, 2, 4)	0.31145588
(5, 6, 7, 0, 4)	(1, 2, 3)	0.42756594
(5, 6, 7, 1, 2)	(0, 3, 4)	0.30064206
(5, 6, 7, 1, 3)	(0, 2, 4)	0.18889328
(5, 6, 7, 1, 4)	(0, 2, 3)	0.31145588
(5, 6, 7, 2, 3)	(0, 1, 4)	0.30064206
(5, 6, 7, 2, 4)	(0, 1, 3)	0.31145588
(5, 6, 7, 3, 4)	(0, 1, 2)	0.42756594
(5, 6, 0, 1, 2)	(7, 3, 4)	0.30064206
(5, 6, 0, 1, 3)	(7, 2, 4)	0.19534582
(5, 6, 0, 1, 4)	(7, 2, 3)	0.30064206
(5, 6, 0, 2, 3)	(7, 1, 4)	0.19534582
(5, 6, 0, 2, 4)	(7, 1, 3)	0.18889328
(5, 6, 0, 3, 4)	(7, 1, 2)	0.31145588
(5, 6, 1, 2, 3)	(7, 0, 4)	0.30064206
(5, 6, 1, 2, 4)	(7, 0, 3)	0.30064206
(5, 6, 1, 3, 4)	(7, 0, 2)	0.31145588
(5, 6, 2, 3, 4)	(7, 0, 1)	0.42756594
(5, 7, 0, 1, 2)	(6, 3, 4)	0.31145588
(5, 7, 0, 1, 3)	(6, 2, 4)	0.18889328
(5, 7, 0, 1, 4)	(6, 2, 3)	0.30064206
(5, 7, 0, 2, 3)	(6, 1, 4)	0.19534582
(5, 7, 0, 2, 4)	(6, 1, 3)	0.19534582
(5, 7, 0, 3, 4)	(6, 1, 2)	0.30064206
(5, 7, 1, 2, 3)	(6, 0, 4)	0.18889328
(5, 7, 1, 2, 4)	(6, 0, 3)	0.19534582
(5, 7, 1, 3, 4)	(6, 0, 2)	0.18889328
(5, 7, 2, 3, 4)	(6, 0, 1)	0.31145588
(5, 0, 1, 2, 3)	(6, 7, 4)	0.31145588
(5, 0, 1, 2, 4)	(6, 7, 3)	0.30064206
(5, 0, 1, 3, 4)	(6, 7, 2)	0.30064206
(5, 0, 2, 3, 4)	(6, 7, 1)	0.31145588
(5, 1, 2, 3, 4)	(6, 7, 0)	0.42756594
(5, 6, 7, 0, 1, 2)	(3, 4)	0.52766357
(5, 6, 7, 0, 1, 3)	(2, 4)	0.29980473
(5, 6, 7, 0, 1, 4)	(2, 3)	0.52766357
(5, 6, 7, 0, 2, 3)	(1, 4)	0.29544345
(5, 6, 7, 0, 2, 4)	(1, 3)	0.29980473
(5, 6, 7, 0, 3, 4)	(1, 2)	0.52766357
(5, 6, 7, 1, 2, 3)	(0, 4)	0.27817710
(5, 6, 7, 1, 2, 4)	(0, 3)	0.29544345
(5, 6, 7, 1, 3, 4)	(0, 2)	0.29980473
(5, 6, 7, 2, 3, 4)	(0, 1)	0.52766357
(5, 6, 0, 1, 2, 3)	(7, 4)	0.29544345
(5, 6, 0, 1, 2, 4)	(7, 3)	0.27817710
(5, 6, 0, 1, 3, 4)	(7, 2)	0.29544345
(5, 6, 0, 2, 3, 4)	(7, 1)	0.29980473
(5, 6, 1, 2, 3, 4)	(7, 0)	0.52766357
(5, 7, 0, 1, 2, 3)	(6, 4)	0.29980473
(5, 7, 0, 1, 2, 4)	(6, 3)	0.29544345
(5, 7, 0, 1, 3, 4)	(6, 2)	0.27817710
(5, 7, 0, 2, 3, 4)	(6, 1)	0.29544345
(5, 7, 1, 2, 3, 4)	(6, 0)	0.29980473
(5, 0, 1, 2, 3, 4)	(6, 7)	0.52766357
(5, 6, 7, 0, 1, 2, 3)	(4,)	0.50000000
(5, 6, 7, 0, 1, 2, 4)	(3,)	0.50000000
(5, 6, 7, 0, 1, 3, 4)	(2,)	0.50000000
(5, 6, 7, 0, 2, 3, 4)	(1,)	0.50000000
(5, 6, 7, 1, 2, 3, 4)	(0,)	0.50000000
(5, 6, 0, 1, 2, 3, 4)	(7,)	0.50000000
(5, 7, 0, 1, 2, 3, 4)	(6,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.97194283
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

- C_exact = 0.97194283
- C_LB = 0.31394240
- Gap = 0.65800042


8-Body RDM (Sites 6,7,0,1,2,3,4,5)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 8-Body RDM (Sites 6,7,0,1,2,3,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000001
Eigenvalue min	-3.314747e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

Convex Roof Analysis
==========

Lower Bounds:
- Paper (witness) = 0.00000000
- Notes (6-9) = 0.31394240
- Best C_LB = 0.31394240

C_GME (unitary optimization - Cayley Transform):
- C_num (unitary opt) = 0.97194283 BEST
- (20 restarts)

C_GME (unitary optimization - Expoenetial Map):
- C_num = 0.97194283 BEST
- (L-BFGS-B, 20 restarts)

Upper Bound:
- C_UB (eigendecomp) = 0.97194283

Status: GME
Bounds consistent: YES
Convergence: TIGHT

GME Analysis
==========

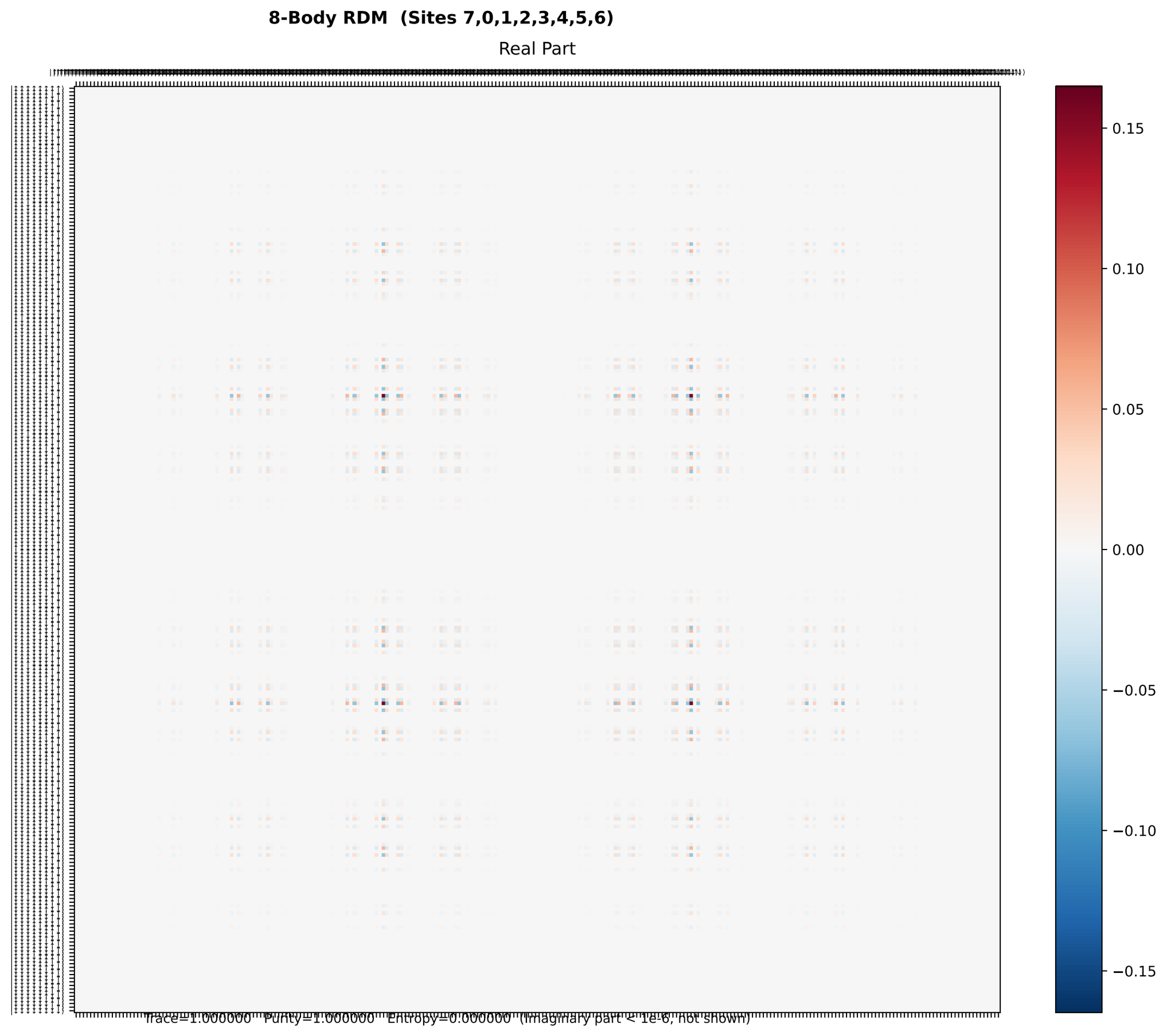
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(6,)	(7, 0, 1, 2, 3, 4, 5)	0.50000000
(6, 7)	(0, 1, 2, 3, 4, 5)	0.52766357
(6, 0)	(7, 1, 2, 3, 4, 5)	0.29980473
(6, 1)	(7, 0, 2, 3, 4, 5)	0.29544345
(6, 2)	(7, 0, 1, 3, 4, 5)	0.27817710
(6, 3)	(7, 0, 1, 2, 4, 5)	0.29544345
(6, 4)	(7, 0, 1, 2, 3, 5)	0.29980473
(6, 5)	(7, 0, 1, 2, 3, 4)	0.52766357
(6, 7, 0)	(1, 2, 3, 4, 5)	0.42756594
(6, 7, 1)	(0, 2, 3, 4, 5)	0.31145588
(6, 7, 2)	(0, 1, 3, 4, 5)	0.30064206
(6, 7, 3)	(0, 1, 2, 4, 5)	0.30064206
(6, 7, 4)	(0, 1, 2, 3, 5)	0.31145588
(6, 7, 5)	(0, 1, 2, 3, 4)	0.42756594
(6, 0, 1)	(7, 2, 3, 4, 5)	0.31145588
(6, 0, 2)	(7, 1, 3, 4, 5)	0.18889328
(6, 0, 3)	(7, 1, 2, 4, 5)	0.19534582
(6, 0, 4)	(7, 1, 2, 3, 5)	0.18889328
(6, 0, 5)	(7, 1, 2, 3, 4)	0.31145588
(6, 1, 2)	(7, 0, 3, 4, 5)	0.30064206
(6, 1, 3)	(7, 0, 2, 4, 5)	0.19534582
(6, 1, 4)	(7, 0, 2, 3, 5)	0.19534582
(6, 1, 5)	(7, 0, 2, 3, 4)	0.30064206
(6, 2, 3)	(7, 0, 1, 4, 5)	0.30064206
(6, 2, 4)	(7, 0, 1, 3, 5)	0.18889328
(6, 2, 5)	(7, 0, 1, 3, 4)	0.30064206
(6, 3, 4)	(7, 0, 1, 2, 5)	0.31145588
(6, 3, 5)	(7, 0, 1, 2, 4)	0.31145588
(6, 4, 5)	(7, 0, 1, 2, 3)	0.42756594
(6, 7, 0, 1)	(2, 3, 4, 5)	0.47865643
(6, 7, 0, 2)	(1, 3, 4, 5)	0.27204245
(6, 7, 0, 3)	(1, 2, 4, 5)	0.29330035
(6, 7, 0, 4)	(1, 2, 3, 5)	0.27204245
(6, 7, 0, 5)	(1, 2, 3, 4)	0.47865643
(6, 7, 1, 2)	(0, 3, 4, 5)	0.29330035
(6, 7, 1, 3)	(0, 2, 4, 5)	0.20203136
(6, 7, 1, 4)	(0, 2, 3, 5)	0.20033735
(6, 7, 1, 5)	(0, 2, 3, 4)	0.27204245
(6, 7, 2, 3)	(0, 1, 4, 5)	0.35325200
(6, 7, 2, 4)	(0, 1, 3, 5)	0.20203136
(6, 7, 2, 5)	(0, 1, 3, 4)	0.29330035
(6, 7, 3, 4)	(0, 1, 2, 5)	0.29330035
(6, 7, 3, 5)	(0, 1, 2, 4)	0.27204245
(6, 7, 4, 5)	(0, 1, 2, 3)	0.47865643
(6, 0, 1, 2)	(7, 3, 4, 5)	0.27204245
(6, 0, 1, 3)	(7, 2, 4, 5)	0.20033735
(6, 0, 1, 4)	(7, 2, 3, 5)	0.20203136
(6, 0, 1, 5)	(7, 2, 3, 4)	0.29330035
(6, 0, 2, 3)	(7, 1, 4, 5)	0.20203136
(6, 0, 2, 4)	(7, 1, 3, 5)	0.13053255
(6, 0, 2, 5)	(7, 1, 3, 4)	0.20203136
(6, 0, 3, 4)	(7, 1, 2, 5)	0.20203136
(6, 0, 3, 5)	(7, 1, 2, 4)	0.20033735
(6, 0, 4, 5)	(7, 1, 2, 3)	0.27204245
(6, 1, 2, 3)	(7, 0, 4, 5)	0.29330035
(6, 1, 2, 4)	(7, 0, 3, 5)	0.20203136
(6, 1, 2, 5)	(7, 0, 3, 4)	0.35325200
(6, 1, 3, 4)	(7, 0, 2, 5)	0.20033735
(6, 1, 3, 5)	(7, 0, 2, 4)	0.20203136
(6, 1, 4, 5)	(7, 0, 2, 3)	0.29330035
(6, 2, 3, 4)	(7, 0, 1, 5)	0.27204245
(6, 2, 3, 5)	(7, 0, 1, 4)	0.29330035
(6, 2, 4, 5)	(7, 0, 1, 3)	0.27204245
(6, 3, 4, 5)	(7, 0, 1, 2)	0.47865643
(6, 7, 0, 1, 2)	(3, 4, 5)	0.42756594
(6, 7, 0, 1, 3)	(2, 4, 5)	0.31145588
(6, 7, 0, 1, 4)	(2, 3, 5)	0.31145588
(6, 7, 0, 1, 5)	(2, 3, 4)	0.42756594
(6, 7, 0, 2, 3)	(1, 4, 5)	0.30064206
(6, 7, 0, 2, 4)	(1, 3, 5)	0.18889328
(6, 7, 0, 2, 5)	(1, 3, 4)	0.31145588
(6, 7, 0, 3, 4)	(1, 2, 5)	0.30064206
(6, 7, 0, 3, 5)	(1, 2, 4)	0.31145588
(6, 7, 0, 4, 5)	(1, 2, 3)	0.42756594
(6, 7, 1, 2, 3)	(0, 4, 5)	0.30064206
(6, 7, 1, 2, 4)	(0, 3, 5)	0.19534582
(6, 7, 1, 2, 5)	(0, 3, 4)	0.30064206
(6, 7, 1, 3, 4)	(0, 2, 5)	0.19534582
(6, 7, 1, 3, 5)	(0, 2, 4)	0.18889328
(6, 7, 1, 4, 5)	(0, 2, 3)	0.31145588
(6, 7, 2, 3, 4)	(0, 1, 5)	0.30064206
(6, 7, 2, 3, 5)	(0, 1, 4)	0.30064206
(6, 7, 2, 4, 5)	(0, 1, 3)	0.31145588
(6, 7, 3, 4, 5)	(0, 1, 2)	0.42756594
(6, 0, 1, 2, 3)	(7, 4, 5)	0.31145588
(6, 0, 1, 2, 4)	(7, 3, 5)	0.18889328
(6, 0, 1, 2, 5)	(7, 3, 4)	0.30064206
(6, 0, 1, 3, 4)	(7, 2, 5)	0.19534582
(6, 0, 1, 3, 5)	(7, 2, 4)	0.19534582
(6, 0, 1, 4, 5)	(7, 2, 3)	0.30064206
(6, 0, 2, 3, 4)	(7, 1, 5)	0.18889328
(6, 0, 2, 3, 5)	(7, 1, 4)	0.19534582
(6, 0, 2, 4, 5)	(7, 1, 3)	0.18889328
(6, 0, 3, 4, 5)	(7, 1, 2)	0.31145588
(6, 1, 2, 3, 4)	(7, 0, 5)	0.31145588
(6, 1, 2, 3, 5)	(7, 0, 4)	0.30064206
(6, 1, 2, 4, 5)	(7, 0, 3)	0.30064206
(6, 1, 3, 4, 5)	(7, 0, 2)	0.31145588
(6, 2, 3, 4, 5)	(7, 0, 1)	0.42756594
(6, 7, 0, 1, 2, 3)	(4, 5)	0.52766357
(6, 7, 0, 1, 2, 4)	(3, 5)	0.29980473
(6, 7, 0, 1, 2, 5)	(3, 4)	0.52766357
(6, 7, 0, 1, 3, 4)	(2, 5)	0.29544345
(6, 7, 0, 1, 3, 5)	(2, 4)	0.29980473
(6, 7, 0, 1, 4, 5)	(2, 3)	0.52766357
(6, 7, 0, 2, 3, 4)	(1, 5)	0.27817710
(6, 7, 0, 2, 3, 5)	(1, 4)	0.29544345
(6, 7, 0, 2, 4, 5)	(1, 3)	0.29980473
(6, 7, 0, 3, 4, 5)	(1, 2)	0.52766357
(6, 7, 1, 2, 3, 4)	(0, 5)	0.29544345
(6, 7, 1, 2, 3, 5)	(0, 4)	0.27817710
(6, 7, 1, 2, 4, 5)	(0, 3)	0.29544345
(6, 7, 1, 3, 4, 5)	(0, 2)	0.29980473
(6, 7, 2, 3, 4, 5)	(0, 1)	0.52766357
(6, 0, 1, 2, 3, 4)	(7, 5)	0.29980473
(6, 0, 1, 2, 3, 5)	(7, 4)	0.29544345
(6, 0, 1, 2, 4, 5)	(7, 3)	0.27817710
(6, 0, 1, 3, 4, 5)	(7, 2)	0.29544345
(6, 0, 2, 3, 4, 5)	(7, 1)	0.29980473
(6, 1, 2, 3, 4, 5)	(7, 0)	0.52766357
(6, 7, 0, 1, 2, 3, 4)	(5,)	0.50000000
(6, 7, 0, 1, 2, 3, 5)	(4,)	0.50000000
(6, 7, 0, 1, 2, 4, 5)	(3,)	0.50000000
(6, 7, 0, 1, 3, 4, 5)	(2,)	0.50000000
(6, 7, 0, 2, 3, 4, 5)	(1,)	0.50000000
(6, 7, 1, 2, 3, 4, 5)	(0,)	0.50000000
(6, 0, 1, 2, 3, 4, 5)	(7,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.97194283
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

- C_exact = 0.97194283
- C_LB = 0.31394240
- Gap = 0.65800042


8-Body RDM (Sites 7,0,1,2,3,4,5,6)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 8-Body RDM (Sites 7,0,1,2,3,4,5,6)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000001
Eigenvalue min	-2.920092e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

Convex Roof Analysis
==========

Lower Bounds:
- Paper (witness) = 0.00000000
- Notes (6-9) = 0.31394240
- Best C_LB = 0.31394240

C_GME (unitary optimization - Cayley Transform):
- C_num (unitary opt) = 0.97194283 BEST
- (20 restarts)

C_GME (unitary optimization - Expoenetial Map):
- C_num = 0.97194283 BEST
- (L-BFGS-B, 20 restarts)

Upper Bound:
- C_UB (eigendecomp) = 0.97194283

Status: GME
Bounds consistent: YES
Convergence: TIGHT

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(7,)	(0, 1, 2, 3, 4, 5, 6)	0.50000000
(7, 0)	(1, 2, 3, 4, 5, 6)	0.52766357
(7, 1)	(0, 2, 3, 4, 5, 6)	0.29980473
(7, 2)	(0, 1, 3, 4, 5, 6)	0.29544345
(7, 3)	(0, 1, 2, 4, 5, 6)	0.27817710
(7, 4)	(0, 1, 2, 3, 5, 6)	0.29544345
(7, 5)	(0, 1, 2, 3, 4, 6)	0.29980473
(7, 6)	(0, 1, 2, 3, 4, 5)	0.52766357
(7, 0, 1)	(2, 3, 4, 5, 6)	0.42756594
(7, 0, 2)	(1, 3, 4, 5, 6)	0.31145588
(7, 0, 3)	(1, 2, 4, 5, 6)	0.30064206
(7, 0, 4)	(1, 2, 3, 5, 6)	0.30064206
(7, 0, 5)	(1, 2, 3, 4, 6)	0.31145588
(7, 0, 6)	(1, 2, 3, 4, 5)	0.42756594
(7, 1, 2)	(0, 3, 4, 5, 6)	0.31145588
(7, 1, 3)	(0, 2, 4, 5, 6)	0.18889328
(7, 1, 4)	(0, 2, 3, 5, 6)	0.19534582
(7, 1, 5)	(0, 2, 3, 4, 6)	0.18889328
(7, 1, 6)	(0, 2, 3, 4, 5)	0.31145588
(7, 2, 3)	(0, 1, 4, 5, 6)	0.30064206
(7, 2, 4)	(0, 1, 3, 5, 6)	0.19534582
(7, 2, 5)	(0, 1, 3, 4, 6)	0.19534582
(7, 2, 6)	(0, 1, 3, 4, 5)	0.30064206
(7, 3, 4)	(0, 1, 2, 5, 6)	0.30064206
(7, 3, 5)	(0, 1, 2, 4, 6)	0.18889328
(7, 3, 6)	(0, 1, 2, 4, 5)	0.30064206
(7, 4, 5)	(0, 1, 2, 3, 6)	0.31145588
(7, 4, 6)	(0, 1, 2, 3, 5)	0.31145588
(7, 5, 6)	(0, 1, 2, 3, 4)	0.42756594
(7, 0, 1, 2)	(3, 4, 5, 6)	0.47865643
(7, 0, 1, 3)	(2, 4, 5, 6)	0.27204245
(7, 0, 1, 4)	(2, 3, 5, 6)	0.29330035
(7, 0, 1, 5)	(2, 3, 4, 6)	0.27204245
(7, 0, 1, 6)	(2, 3, 4, 5)	0.47865643
(7, 0, 2, 3)	(1, 4, 5, 6)	0.29330035
(7, 0, 2, 4)	(1, 3, 5, 6)	0.20203136
(7, 0, 2, 5)	(1, 3, 4, 6)	0.20033735
(7, 0, 2, 6)	(1, 3, 4, 5)	0.27204245
(7, 0, 3, 4)	(1, 2, 5, 6)	0.35325200
(7, 0, 3, 5)	(1, 2, 4, 6)	0.20203136
(7, 0, 3, 6)	(1, 2, 4, 5)	0.29330035
(7, 0, 4, 5)	(1, 2, 3, 6)	0.29330035
(7, 0, 4, 6)	(1, 2, 3, 5)	0.27204245
(7, 0, 5, 6)	(1, 2, 3, 4)	0.47865643
(7, 1, 2, 3)	(0, 4, 5, 6)	0.27204245
(7, 1, 2, 4)	(0, 3, 5, 6)	0.20033735
(7, 1, 2, 5)	(0, 3, 4, 6)	0.20203136
(7, 1, 2, 6)	(0, 3, 4, 5)	0.29330035
(7, 1, 3, 4)	(0, 2, 5, 6)	0.20203136
(7, 1, 3, 5)	(0, 2, 4, 6)	0.13053255
(7, 1, 3, 6)	(0, 2, 4, 5)	0.20203136
(7, 1, 4, 5)	(0, 2, 3, 6)	0.20203136
(7, 1, 4, 6)	(0, 2, 3, 5)	0.20033735
(7, 1, 5, 6)	(0, 2, 3, 4)	0.27204245
(7, 2, 3, 4)	(0, 1, 5, 6)	0.29330035
(7, 2, 3, 5)	(0, 1, 4, 6)	0.20203136
(7, 2, 3, 6)	(0, 1, 4, 5)	0.35325200
(7, 2, 4, 5)	(0, 1, 3, 6)	0.20033735
(7, 2, 4, 6)	(0, 1, 3, 5)	0.20203136
(7, 2, 5, 6)	(0, 1, 3, 4)	0.29330035
(7, 3, 4, 5)	(0, 1, 2, 6)	0.27204245
(7, 3, 4, 6)	(0, 1, 2, 5)	0.29330035
(7, 3, 5, 6)	(0, 1, 2, 4)	0.27204245
(7, 4, 5, 6)	(0, 1, 2, 3)	0.47865643
(7, 0, 1, 2, 3)	(4, 5, 6)	0.42756594
(7, 0, 1, 2, 4)	(3, 5, 6)	0.31145588
(7, 0, 1, 2, 5)	(3, 4, 6)	0.31145588
(7, 0, 1, 2, 6)	(3, 4, 5)	0.42756594
(7, 0, 1, 3, 4)	(2, 5, 6)	0.30064206
(7, 0, 1, 3, 5)	(2, 4, 6)	0.18889328
(7, 0, 1, 3, 6)	(2, 4, 5)	0.31145588
(7, 0, 1, 4, 5)	(2, 3, 6)	0.30064206
(7, 0, 1, 4, 6)	(2, 3, 5)	0.31145588
(7, 0, 1, 5, 6)	(2, 3, 4)	0.42756594
(7, 0, 2, 3, 4)	(1, 5, 6)	0.30064206
(7, 0, 2, 3, 5)	(1, 4, 6)	0.19534582
(7, 0, 2, 3, 6)	(1, 4, 5)	0.30064206
(7, 0, 2, 4, 5)	(1, 3, 6)	0.19534582
(7, 0, 2, 4, 6)	(1, 3, 5)	0.18889328
(7, 0, 2, 5, 6)	(1, 3, 4)	0.31145588
(7, 0, 3, 4, 5)	(1, 2, 6)	0.30064206
(7, 0, 3, 4, 6)	(1, 2, 5)	0.30064206
(7, 0, 3, 5, 6)	(1, 2, 4)	0.31145588
(7, 0, 4, 5, 6)	(1, 2, 3)	0.42756594
(7, 1, 2, 3, 4)	(0, 5, 6)	0.31145588
(7, 1, 2, 3, 5)	(0, 4, 6)	0.18889328
(7, 1, 2, 3, 6)	(0, 4, 5)	0.30064206
(7, 1, 2, 4, 5)	(0, 3, 6)	0.19534582
(7, 1, 2, 4, 6)	(0, 3, 5)	0.19534582
(7, 1, 2, 5, 6)	(0, 3, 4)	0.30064206
(7, 1, 3, 4, 5)	(0, 2, 6)	0.18889328
(7, 1, 3, 4, 6)	(0, 2, 5)	0.19534582
(7, 1, 3, 5, 6)	(0, 2, 4)	0.18889328
(7, 1, 4, 5, 6)	(0, 2, 3)	0.31145588
(7, 2, 3, 4, 5)	(0, 1, 6)	0.31145588
(7, 2, 3, 4, 6)	(0, 1, 5)	0.30064206
(7, 2, 3, 5, 6)	(0, 1, 4)	0.30064206
(7, 2, 4, 5, 6)	(0, 1, 3)	0.31145588
(7, 3, 4, 5, 6)	(0, 1, 2)	0.42756594
(7, 0, 1, 2, 3, 4)	(5, 6)	0.52766357
(7, 0, 1, 2, 3, 5)	(4, 6)	0.29980473
(7, 0, 1, 2, 3, 6)	(4, 5)	0.52766357
(7, 0, 1, 2, 4, 5)	(3, 6)	0.29544345
(7, 0, 1, 2, 4, 6)	(3, 5)	0.29980473
(7, 0, 1, 2, 5, 6)	(3, 4)	0.52766357
(7, 0, 1, 3, 4, 5)	(2, 6)	0.27817710
(7, 0, 1, 3, 4, 6)	(2, 5)	0.29544345
(7, 0, 1, 3, 5, 6)	(2, 4)	0.29980473
(7, 0, 1, 4, 5, 6)	(2, 3)	0.52766357
(7, 0, 2, 3, 4, 5)	(1, 6)	0.29544345
(7, 0, 2, 3, 4, 6)	(1, 5)	0.27817710
(7, 0, 2, 3, 5, 6)	(1, 4)	0.29544345
(7, 0, 2, 4, 5, 6)	(1, 3)	0.29980473
(7, 0, 3, 4, 5, 6)	(1, 2)	0.52766357
(7, 1, 2, 3, 4, 5)	(0, 6)	0.29980473
(7, 1, 2, 3, 4, 6)	(0, 5)	0.29544345
(7, 1, 2, 3, 5, 6)	(0, 4)	0.27817710
(7, 1, 2, 4, 5, 6)	(0, 3)	0.29544345
(7, 1, 3, 4, 5, 6)	(0, 2)	0.29980473
(7, 2, 3, 4, 5, 6)	(0, 1)	0.52766357
(7, 0, 1, 2, 3, 4, 5)	(6,)	0.50000000
(7, 0, 1, 2, 3, 4, 6)	(5,)	0.50000000
(7, 0, 1, 2, 3, 5, 6)	(4,)	0.50000000
(7, 0, 1, 2, 4, 5, 6)	(3,)	0.50000000
(7, 0, 1, 3, 4, 5, 6)	(2,)	0.50000000
(7, 0, 2, 3, 4, 5, 6)	(1,)	0.50000000
(7, 1, 2, 3, 4, 5, 6)	(0,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.97194283
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

C_exact = 0.97194283
C_LB = 0.31394240
Gap = 0.65800042